



Project Report On

Multimodal Emotion Recognition Using Audio, Video and Text Fusion

*Submitted in partial fulfillment of the requirements for the
award of the degree of*

Bachelor of Technology

in

Computer Science and Engineering

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March 2026

CERTIFICATE

*This is to certify that the project report entitled “**Multimodal Emotion Recognition Using Audio, Video and Text Fusion**” is a bonafide record of the work done by **Savio Tomy(U2203191)**, **V Karthik(U2203206)**, **Volwin Shaji(U2203213)**, **Yuhan John(U2203215)** submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in “Computer Science and Engineering” during the academic year 2025-2026.*

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ACKNOWLEDGMENT

We wish to express our sincere gratitude towards **Rev. Dr. Jaison Paul Mulerikkal CMI**, Principal of RSET, and **Dr. Preetha K.G**, Professor ,Head of the Department of Computer Science and Engineering, for providing us with the opportunity to undertake our project, ” Multimodal Emotion Recognition Using Audio , Video and Text Fusion”.

We are highly indebted to our project coordinator, **Ms. Jisha Mary Jose**, Assistant Professor, Dept. of CSE, for her valuable support.

It is indeed our pleasure and a moment of satisfaction for us to express our sincere gratitude to our project guide **Dr. Preetha K.G** , Professor, Dept. of CSE,for her patience and all the priceless advice and wisdom she has shared with us.

Last but not the least, We would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

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Abstract

Emotion recognition is an important area in human–computer interaction and affective computing, enabling machines to respond intelligently in applications such as virtual assistants, healthcare systems, and customer support. Traditional methods rely on a single modality like facial expressions or speech, which often results in incomplete understanding since human emotions are multidimensional. This project proposes a multimodal emotion recognition system that integrates visual, acoustic, and contextual information to improve accuracy and reliability. The system processes audiovisual inputs, where video frames capture facial expressions and temporal changes, while audio signals provide speech characteristics such as pitch, tone, and intensity. Contextual understanding is incorporated through descriptive representations derived from observed behavioral patterns, enabling a more comprehensive interpretation of emotions.

The methodology involves multimodal data acquisition followed by preprocessing, including normalization of video frames and conversion of audio into Mel spectrograms. A 3D-CNN extracts spatial and temporal features from facial expressions, while a 2D-CNN captures acoustic features from speech signals. A contextual module enhances interpretability by generating descriptive insights from facial and vocal patterns. These features are fused using a transformer-based model to learn cross-modal relationships and produce a unified emotional representation. A final classification layer predicts the emotion along with a confidence score. Experimental evaluation on the RAVDESS dataset demonstrates improved accuracy and robustness compared to unimodal approaches. Overall, the system highlights the effectiveness of combining multimodal analysis with advanced fusion techniques to develop accurate and context-aware emotion recognition systems for real-world applications.

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List of Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
3D-CNN	Three-Dimensional Convolutional Neural Network
DL	Deep Learning
ML	Machine Learning
RNN	Recurrent Neural Network
LLM	Large Language Model
FAN	Facial Action Network
FACS	Facial Action Coding System
SER	Speech Emotion Recognition
MER	Multimodal Emotion Recognition
MFCC	Mel-Frequency Cepstral Coefficients
Mel	Mel Frequency Scale
STFT	Short-Time Fourier Transform
ASR	Automatic Speech Recognition
GPU	Graphics Processing Unit
CPU	Central Processing Unit
API	Application Programming Interface
GUI	Graphical User Interface
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
FER	Facial Expression Recognition
HCI	Human Computer Interaction
NLP	Natural Language Processing

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Chapter 1

Introduction

This chapter presents the fundamental ideas, guiding principles, and main driving force behind the project "Multimodal Emotion Recognition Using Audio, Video, and Text Fusion." It highlights the urgent need to create a more intelligent, context-aware framework for affective computing, which is the study and creation of systems that are capable of identifying, interpreting, processing, and simulating human emotions. Additionally, it describes the intrinsic difficulties and performance bottlenecks that conventional single-modality systems encounter and presents the cutting-edge deep learning techniques used to overcome these constraints. The problem context, system goals, architectural scope, and overall significance of this work in the quickly developing field of intelligent human-computer interaction are all thoroughly described in the following sections.

1.1 Background

In order for machines to comprehend and react to human emotions in a natural, sympathetic, and intuitive way, emotion recognition is a crucial and fundamental aspect of human-computer interaction [1]. Artificial intelligence's quick development has created revolutionary opportunities in a number of fields, necessitating the development of computational systems that can precisely determine user sentiment and modify their responses accordingly [2]. However, conventional systems that only use one modality such as standalone video processing, isolated audio analysis, or purely text-based sentiment analysis frequently fall short of capturing the entire emotional context, which can result in serious errors and system failures in practical applications [3, 4].

An emotional state is rarely expressed through a single, isolated channel in natural human communication. Rather, it is a synchronized and intricate synthesis of both verbal and nonverbal cues [4]. Even though the current conventional models are well-researched and optimized in their particular, limited domains, they consistently perform poorly when used in complex, unrestricted environments [5]. For instance, a passive-aggressive or sarcastic transcript could be

entirely misinterpreted by a text-based system, but an integrated human-like system could readily identify the underlying sarcasm through subtle facial micro-expressions and shifting vocal tones. Our suggested system attempts to more closely resemble human psychological perception by methodically merging various sensory data sources [6]. This multisensory, integrated approach is essential for opening up cutting-edge, incredibly dependable applications in interactive entertainment, adaptive education platforms, and healthcare diagnostics [1, 2].

1.2 Problem Definition

The majority of emotion recognition systems currently in use in academia and industry are unimodal, mainly depending on written text, vocal speech patterns, or facial expressions. Their practical efficacy is intrinsically limited by this isolated approach. These isolated systems constantly fail to correctly interpret emotions in real-world situations, where vocal tone, subtle behavioral cues, and environmental context are crucial. This is because human emotional expression is inherently complex and frequently depends on contradictory signals, such as crying tears of joy or smiling through frustration.

When a particular modality is hampered by environmental factors like background noise, dim ambient lighting, or unclear language phrasing, relying solely on one data stream will unavoidably result in serious misclassifications [7]. Such misclassifications can have negative consequences, from misinterpreting a patient’s psychological state during a telehealth session to annoying a customer interacting with a service bot [2]. In order to provide a fundamentally more accurate, resilient, and context-aware emotion detection framework, the main goal of this research is to design and develop a robust multimodal emotion recognition system that mathematically fuses audio, visual, and textual features [8].

1.3 Scope and Motivation

This project’s scope includes designing, implementing, and rigorously evaluating a full multimodal deep learning architecture. For a thorough, synchronized emotion analysis, the suggested system takes video input that includes spoken speech and synchronized audio tracks. To maintain strict data consistency across all channels, the technical scope includes text transcription, video frame extraction, and audio signal preprocessing. Highly specialized neural networks a Two-Dimensional Convolutional Neural Network for processing audio Mel Spectrogram images and a Three-Dimensional Convolutional Neural Network for spatial and temporal video analysispower feature extraction.

Additionally, the system has a special module that uses particular facial action units and audio prosody metrics like pitch and loudness to generate rule-based contextual descriptions . Lastly, the framework uses a multimodal fusion technique based on Large Language Models to smoothly integrate contextual qualitative linguistic information with deep quantitative mathematical features . The benchmark Ryerson Audio-Visual Database of Emotional Speech and Song is used to rigorously assess the overall performance.

The fundamental driving force is the rapidly increasing need for emotion-aware AI in sensitive and time-sensitive applications where human-machine misinterpretation is completely intolerable . Automated mental health monitoring, emotion-aware AI in remote learning, improved HCI, and advanced entertainment analytics all require accurate, fail-safe emotion detection . The computational flexibility and thorough situational awareness required to manage these real-world deployments dependably are simply absent from current single-modality approaches [?]

1.4 Objectives

- To design and implement a robust multimodal emotion recognition system that seamlessly fuses audio, visual, and textual data features to achieve superior classification accuracy compared to single-modality baseline.
- To process synchronized video input by extracting deep emotional features using a Three-Dimensional Convolutional Neural Network architecture for visual frames [9] and a Two-Dimensional Convolutional Neural Network for processing acoustic signals into spectrograms [7].
- To develop a unique rule-based context generation module capable of extracting distinct physical facial action units and audio prosody elements to form descriptive textual cues for the model [10].
- To implement an advanced Large Language Model based mathematical fusion mechanism that effectively and efficiently combines deep multidimensional numerical features with semantic context descriptions without losing critical emotional data [6, 11].
- To generate final emotion classifications across standard emotional categories, accompanied by reliable mathematical confidence scores for user transparency.
- To rigorously evaluate the system’s performance, precision, recall, and generalizability using standardized benchmark datasets such as the Ryerson Audio-Visual Database of Emotional Speech and Song [5, 12].

1.5 Challenges

Even with the deployment of advanced deep learning architectures, several real-world challenges heavily complicate the reliable deployment of multimodal emotion recognition network. Multimodal complexity and the heterogeneous nature of the data represent the primary obstacles; combining continuous audio frequencies, spatial-temporal video pixels, and discrete textual tokens introduces strict synchronization and alignment challenges, where even minor timing offsets can lead to inconsistent and erroneous emotion recognition. Furthermore, accuracy issues remain a pervasive concern in unconstrained, real-world environments, as mathematical fusion models may still produce false predictions or misclassify subtle emotions under noisy, poorly lit, or highly ambiguous conditions.

From a system performance and hardware perspective, real-time computational constraints pose a significant hurdle. Processing three dense, high-dimensional input modalities simultaneously is extremely computationally expensive [13]. This high processing overhead may cause unacceptable system latency or require highly specialized, expensive hardware arrays that limit consumer accessibility. Finally, ensuring long-term scalability and software maintenance requires continuous model updates and intensive retraining procedures to maintain high accuracy as new datasets, evolving slang, and varying demographic expressions emerge over time [2, 4].

1.6 Assumptions

The following foundational assumptions are made in the design, training, and evaluation phases of the proposed system:

- Users interacting with the system will provide relatively clear video and audio recordings with minimal environmental background noise to ensure baseline sensory data quality.
- Human facial expressions must be clearly visible and unobstructed within the camera frame to allow for accurate Action Unit extraction [10], and microphone audio quality must be sufficient for reliable prosody analysis [9].
- When applicable, users respond to the textual input prompts or initial questionnaires honestly and completely.
- The software system operates under adequate computational resources, specifically within a Graphics Processing Unit enabled hardware environment capable of handling simultaneous deep feature extraction and Large Language Model inference without memory overflow.

- The emotional states expressed by the users during testing belong strictly to the predefined emotion categories (such as angry, happy, sad, fearful, surprise, disgust, calm, and neutral) utilized during the model’s initial training phase [5].

1.7 Societal and Industrial Relevance

The proposed technological framework has direct, immediate, and highly impactful applicability across multiple global sectors [1]. In the field of mental health monitoring and telehealth, it provides clinical professionals with an automated, objective, and continuous tool to track patient emotional states over time, potentially identifying depressive or anxious trends before they escalate [12]. In the educational sector, emotion-aware Artificial Intelligence can adapt teaching methodologies and pacing in real-time based on monitoring student engagement, confusion, and frustration levels, thereby personalizing the learning experience [2].

Industrially, this research successfully bridges the often-difficult gap between theoretical academic exploration and practical, commercial software deployment. This significant achievement is evidenced by the successful development of this system as an industry-supported project in direct collaboration with BaseOne Innovations Private Limited, located in Kakkanad. This partnership directly integrates real-world commercial software requirements, scalability concerns, and user experience methodologies with advanced Artificial Intelligence research [8].

1.8 Organization of the Report

The remainder of this document is organized as follows: Chapter 1 provides a comprehensive overview of the project, including the background, problem definition, implementation scope, objectives, challenges, assumptions, and societal relevance. Chapter 2 presents a detailed literature survey analyzing existing approaches to emotion recognition, heavily focusing on the strengths and limitations of current feature fusion models [3, 4, 1]. Chapter 3 describes the overarching architecture diagram, sequence diagrams, and detailed software module divisions including the Deep Feature Extraction Stream and the Rule-Based Context Generation Stream. Chapter 4 provides a highly detailed technical description of the hardware constraints and software implementation methodologies. Chapter 5 presents the empirical evaluation results, including confusion matrices and performance metrics tested against the Ryerson Audio-Visual Database of Emotional Speech and Song. Finally, Chapter 6 concludes the report and outlines future

plans for real-world application deployment.

1.9 Summary of Chapter

This chapter thoroughly introduced the motivating problem of enabling accurate, human-like, and context-aware emotion recognition through the mathematical fusion of multiple data modalities . The severe limitations of conventional single-modality approaches in dynamic, real-world environments were discussed in detail . Furthermore, the overall scope, specific objectives, anticipated technical challenges, and significant industrial contributions of the proposed Artificial Intelligence framework were outlined to provide a clear roadmap for the subsequent technical chapters.

Chapter 2

Literature Survey

Understanding human emotions using computer systems has become an important research topic in recent years. Emotion recognition plays a significant role in applications such as human-computer interaction, virtual assistants, mental health monitoring, and social robotics. Traditionally, emotion detection systems relied on a single source of information such as facial expressions, speech signals, or text. While these approaches can capture certain emotional cues, they often fail to represent the complete emotional state of a person. Human emotions are usually expressed through a combination of facial movements, tone of voice, and spoken words. Because of this, researchers have increasingly focused on multimodal emotion recognition systems that combine multiple sources of information.

Recent advances in deep learning have significantly improved the ability of machines to analyze complex emotional signals. By combining information from different modalities such as audio, video, and text, these systems are able to produce more reliable emotion predictions. Several research works have proposed different architectures and fusion techniques to improve the performance of multimodal emotion recognition systems. This section reviews the key studies related to this field and draws comparisons across their methodologies, contributions, and limitations.

2.1 Attention-Based Multimodal Emotion Recognition

Problem Addressed

Traditional speech emotion recognition systems often struggle to capture both the spatial characteristics and temporal dynamics of speech signals simultaneously. Existing approaches fail to adequately model the sequential nature of emotional cues embedded in speech, limiting their ability to accurately identify affective states from audio inputs.

Objective

The objective of this work is to develop a 3-D Convolutional Recurrent Neural Network combined with an attention model capable of performing robust speech emotion recognition by jointly

capturing spatial, temporal, and emotionally salient features from speech signals.

Methodology

- **Signal Processing:** Raw speech signals are preprocessed and structured as input representations suitable for three-dimensional convolutional operations, enabling the model to operate over both frequency and time dimensions.
- **3D Convolutional Feature Extraction:** Three-dimensional convolutional layers are applied to the input to simultaneously capture spatial patterns and short-term temporal dynamics present in the speech signal.
- **Recurrent Modeling:** Recurrent layers process the output of the convolutional stages to model long-range temporal dependencies across the speech sequence, preserving contextual emotional information over time.
- **Attention Mechanism:** An attention model is incorporated to assign greater weight to the most emotionally relevant segments of the input, allowing the system to focus on discriminative regions within the speech.
- **Emotion Classification:** The attended feature representations are passed to a classification layer that predicts the emotional category of the input speech sample.

Advantages

The combined use of 3D convolution and recurrent layers enables simultaneous spatial and temporal feature modeling, while the attention mechanism improves the interpretability and discriminative power of the system. Strong performance was demonstrated on the RAVDESS dataset, highlighting the effectiveness of temporal modeling for speech-based emotion recognition.

Disadvantages

The model requires large amounts of labeled training data to generalize effectively. Additionally, the complex multi-stage architecture demands significant computational resources, which may limit its applicability in resource-constrained or real-time deployment environments.

2.2 Emotion Recognition Using Speech and Facial Expressions

Problem Addressed

Single-modality emotion recognition systems are often insufficient for accurately capturing the full range of human affective states. Relying solely on either speech or facial expressions limits the

system’s ability to model the complementary emotional cues present across different modalities, reducing overall recognition accuracy.

Objective

The objective of this work is to develop a multimodal emotion recognition system that effectively combines speech and facial expression modalities using heterogeneous convolutional neural networks and Multimodal Factorized Bilinear pooling to achieve improved emotion classification over single-modality approaches.

Methodology

- **Speech Processing:** Raw speech signals are converted into spectrogram representations, which encode frequency and temporal information. These spectrograms are then analyzed using a dedicated CNN model to extract discriminative acoustic features.
- **Facial Expression Processing:** Facial expression data is processed through a separate CNN stream that extracts visual features corresponding to emotional states from video frames.
- **Multimodal Fusion:** The Multimodal Factorized Bilinear pooling technique is applied to combine the speech and facial feature representations, enabling the model to capture rich cross-modal interactions between the two modalities.
- **Emotion Classification:** The fused multimodal features are passed to a classification layer that predicts the emotional category of the input, leveraging the complementary information from both modalities.

Advantages

The combination of heterogeneous CNNs with MFB pooling enables effective fusion of complementary speech and facial cues, resulting in improved emotion recognition performance over single-modality baselines. The approach was validated on benchmark datasets, demonstrating its generalizability and robustness across different emotional categories.

Disadvantages

The system is sensitive to noisy audio or degraded video quality, which can significantly reduce the effectiveness of the multimodal fusion process. This dependency on clean input signals may limit the system’s reliability in real-world environments where both audio and visual data are subject to corruption.

2.3 Deep Learning Approaches for Multimodal Emotion Detection

Problem Addressed

Traditional machine learning approaches for emotion recognition are limited in their ability to capture the complex spatial and temporal patterns present in multimodal data. Single-stream models fail to jointly exploit the complementary information available from facial expressions and speech signals, resulting in suboptimal classification performance.

Objective

The objective of this work is to explore deep learning architectures that combine convolutional and recurrent neural networks to jointly model spatial and temporal features from facial and speech modalities, achieving improved emotion recognition over conventional machine learning baselines.

Methodology

- **Facial Feature Extraction:** CNN models are applied to facial images to extract spatial features representing visual emotional cues across different expression categories.
- **Temporal Speech Modeling:** LSTM-based models process speech signals to capture the temporal dynamics and sequential patterns associated with emotional states over time.
- **Multimodal Representation Learning:** The spatial features from the CNN stream and the temporal features from the LSTM stream are combined into a unified representation that encodes complementary emotional information from both modalities.
- **Emotion Classification:** The combined multimodal representation is used to classify emotional states, leveraging the strengths of both convolutional and recurrent architectures.

Advantages

The integration of CNN and LSTM models enables simultaneous modeling of spatial and temporal emotional cues, leading to improved classification performance compared with traditional machine learning approaches. The architecture demonstrates the effectiveness of combining complementary deep learning paradigms for multimodal emotion recognition.

Disadvantages

The model is computationally intensive and requires careful hyperparameter tuning to achieve good generalization across different datasets and emotional categories.

Problem Addressed

Detecting depression and recognizing emotions from a single modality is insufficient for capturing the full behavioral complexity of affective states. Systems that rely on only speech or facial expressions overlook the rich contextual and linguistic cues present in language, limiting their diagnostic and recognition capabilities.

Objective

The objective of this work is to apply multimodal deep learning to detect depression and recognize emotions by jointly analyzing speech, facial expressions, and language features using the IEMOCAP dataset, combining deep neural networks with handcrafted feature extraction for comprehensive affective modeling.

Methodology

- **Speech Analysis:** Deep neural networks process speech signals to extract acoustic features that encode prosodic and tonal emotional indicators.
- **Facial Expression Analysis:** Facial expression data is analyzed to capture visual behavioral cues associated with emotional and depressive states.
- **Language Feature Extraction:** Linguistic features are extracted from transcribed speech to model the semantic and contextual dimensions of emotional expression.
- **Hybrid Feature Integration:** A hybrid strategy combining deep learning representations with handcrafted feature extraction is employed to capture both high-level emotional patterns and domain-specific behavioral indicators across all three modalities.
- **Emotion and Depression Classification:** The integrated multimodal features are passed to a classification layer to jointly predict emotional categories and detect indicators of depression.

Advantages

The hybrid approach combining deep learning with handcrafted features allows the system to capture both abstract emotional representations and domain-specific behavioral cues. Jointly modeling speech, facial expressions, and language yields a richer and more comprehensive affective representation than any single modality alone.

Disadvantages

The performance of the system depends heavily on the quality and size of the labeled training data, which may be difficult to obtain at scale for depression-related affective computing tasks.

2.4 LLM-Based and Transformer-Based Multimodal Emotion Fusion

Problem Addressed

Fusing heterogeneous modalities such as audio and visual signals for emotion recognition is a challenging task due to the fundamental differences in their data representations. Conventional fusion methods struggle to create a unified framework that effectively bridges the semantic gap between these diverse input types.

Objective

The objective of this work is to develop a multimodal emotion recognition framework that leverages Large Language Models for feature fusion, enabling a unified and flexible approach to combining emotional cues extracted from audio and visual modalities through natural language representations.

Methodology

- **Multimodal Feature Extraction:** Emotional cues are independently extracted from audio and visual input streams, capturing the affective information present in each modality.
- **Modality-to-Text Conversion:** The extracted audio and visual emotional features are converted into structured textual descriptions, translating heterogeneous modality-specific representations into a common natural language format.
- **LLM-Based Fusion:** The textual descriptions of both modalities are processed by a Large Language Model, which leverages its language understanding capabilities to integrate the multimodal emotional cues and reason over the combined representation.
- **Emotion Prediction:** The LLM produces the final emotion prediction based on the fused natural language representation, unifying the contributions of both the audio and visual modalities within a single inference step.

Advantages

The use of a Large Language Model as a fusion mechanism provides a unified and flexible framework for combining heterogeneous modalities through natural language, eliminating the need for modality-specific fusion architectures. This approach leverages the powerful contextual reasoning capabilities of LLMs to integrate emotional information across diverse input types.

Disadvantages

The conversion of audio and visual signals into textual descriptions may result in the loss of

subtle low-level emotional details that are inherently difficult to capture in natural language form, potentially limiting the system’s sensitivity to fine-grained affective cues.

2.5 Survey and Review Studies

Pan et al. [3] provided a comprehensive review of multimodal emotion recognition covering datasets, preprocessing techniques, feature extraction methods, and fusion strategies. The survey highlights that while significant progress has been made, challenges remain in handling data imbalance, cross-corpus generalization, and real-time deployment.

Zhang et al. [2] conducted a systematic review of deep learning-based multimodal emotion recognition from audio, visual, and text modalities. The review categorizes existing methods by fusion strategy and modality combination, and identifies future research directions including the need for larger and more diverse datasets and more efficient fusion architectures.

Lian et al. [4] surveyed deep learning approaches for multimodal emotion recognition focusing on speech, text, and face modalities. The paper discusses the role of each modality in conveying emotional information and reviews the strengths and limitations of existing fusion approaches.

Wang et al. [11] proposed a Multimodal Transformer Augmented Fusion approach for speech emotion recognition. The system uses transformer-based self-attention to model the relationships between audio features and other contextual signals. The transformer architecture allows the model to dynamically weight the contribution of different features. The method achieved competitive results but requires careful design of the fusion strategy to avoid information loss across modalities.

Ahmed et al. [1] presented a systematic survey on multimodal emotion recognition using various learning algorithms. The study reviews both traditional and deep learning methods and discusses how different modalities contribute to improving overall recognition accuracy.

Ming et al. [10] presented a CNN-LSTM facial expression recognition method that incorporates a two-layer attention mechanism. The first attention layer focuses on spatially important regions within each frame, while the second layer focuses on the most emotionally expressive frames across the temporal sequence. This hierarchical attention strategy improved recognition accuracy on facial expression datasets. However, the method focuses solely on the visual modality and does not incorporate audio or text information.

Le et al. [13] addressed multi-label multimodal emotion recognition using transformer-based fusion combined with emotion-level representation learning. Rather than predicting a single emotion label, the system assigns multiple emotional states to each sample, which better reflects

the complexity of real human emotions. The model showed strong performance on multi-label emotion datasets, although the multi-label setting introduces additional complexity in training.

Khan et al. [8] introduced MemoCMT, a cross-modal transformer-based framework for multimodal emotion recognition. The system uses cross-modal attention to capture interactions between audio, visual, and text modalities simultaneously. This method improves the alignment and integration of features from different sources. The MemoCMT framework demonstrated state-of-the-art results on standard benchmarks, though it requires substantial computational resources due to the complexity of cross-modal attention.

2.6 Research Gap

Although significant progress has been made in multimodal emotion recognition, several challenges still remain [3, 1]. Many existing systems require large datasets and powerful computing resources, which can limit their practical deployment [2]. Most deep learning models operate as black boxes, making it difficult to interpret their predictions in a meaningful way [4]. Furthermore, while transformer-based methods have shown strong results [11, 8, 13], they often struggle to combine low-level perceptual features with high-level semantic information in an interpretable manner. As a result, developing efficient multimodal fusion strategies that integrate audio, visual, and textual features while maintaining both accuracy and interpretability remains an active and open research problem.

2.7 Summary of Chapter

Table 2.1 presents a comparative summary of all the reviewed research works along with their modalities, methods, main contributions, and limitations.

Table 2.1: Comparative Summary of Literature on Multimodal Emotion Recognition

No.	Author(s)	Modalities	Contribution	Limitations
1	Chen et al. [9] (2018)	Audio	Captures temporal speech emotion patterns using 3D-CRNN with attention	Requires large datasets and high compute
2	Zhang et al. [7] (2022)	Audio, Video	Combines audio-visual cues using heterogeneous CNN with MFB pooling	Sensitive to noise and poor video quality
3	Ranganathan et al. [5] (2016)	Audio, Video	Improves emotion classification using CNN+LSTM spatial-temporal fusion	Computationally intensive
4	Tripathi et al. [12] (2018)	Audio, Video, Text	Detects depression-related patterns using deep learning on IEMOCAP	Depends on quality labeled data
5	McEwan et al. [6] (2024)	Audio, Video, Text	Unified multimodal framework for emotion understanding via LLM fusion	Loss of subtle cues in text conversion
6	Pan et al. [3] (2023)	Survey	Comprehensive overview of datasets, features, and fusion methods	Review only, no model proposed
7	Zhang et al. [2] (2024)	Survey	Categorizes DL fusion strategies and identifies future directions	Review only, no model proposed
8	Lian et al. [4] (2023)	Audio, Video, Text	Reviews role of speech, text, and face in emotion recognition	Review only, no model proposed
9	Wang et al. [11] (2023)	Audio	Dynamic feature weighting via transformer augmented fusion	Complex fusion, limited to audio
10	Ahmed et al. [1] (2023)	Survey	Reviews traditional and DL approaches for multimodal emotion recognition	Review only, no model proposed
11	Ming et al. [10] (2022)	Video	Hierarchical two-layer attention in CNN-LSTM for facial recognition	Visual modality only
12	Le et al. [13] (2023)	Audio, Video, Text	Handles multiple simultaneous emotions via transformer multi-label fusion	Increased training complexity
13	Khan et al. [8] (2025)	Audio, Video, Text	State-of-the-art cross-modal alignment using MemoCMT transformer	High computational cost

Chapter 3

Proposed Methodology

The suggested methodology suggests a hybrid multimodal sensory of emotion recognition and tries to detect human emotional states based on the combination of information of various sources, such as visual cues, acoustic patterns, and contextual text data. Human emotions are complicated states of the human psychic life that is manifested by the means of behavioral and physiological indicators. These cues can be facial expressions, voice, patterns of speech and language in context. Single modality systems cannot be effective at capturing the entirety of the emotional context since emotional expressions do not necessarily manifest evidently through a single communication medium. An example is that an individual can have a noncommittal expression on his face yet his voice can have stress or frustration. In the same way, the contextual information revealed with the help of language can convey such emotions that can not be determined with the help of visual or audio signals.

In order to deal with such limitations, the proposed framework uses a multimodal technique that incorporates visual, audio, and textual information as one architecture. The system has multimodal fusion with feature extraction using deep neural networks, rule-based contextual reasoning, and transformer-based multimodal fusion. This mixture design enables the system to take a low-level perceptual and a high-level semantic interpretation of an emotional expression. Deep learning models extract contextual and artistic representations containing rich representations of raw audiovisual cues, whereas the rule-based modules create contextual descriptions that have interpretable semantics to the learned features. These representations are then combined in a transformer-based language model which then carries out the emotion classification.

The suggested approach is a train of processing steps with multiple phases, such as multimodal data capture, data processing, deep features extraction of visual and audio streams, context development based on rules, multimodal fusion with a transformer model, and the ultimate category. All these processes play a role in the conversion of uncoded sensory data to meaningful affective forecasts.

3.1 Multimodal Data: Data Acquisition

Multisensory inputs involve more than just audio and visual information. The acquisition of multimodal input data is a part of the initial stage of the proposed system. A video recording is the main input to the system, which comprises of visual frames and audio synchronous signals. Video recordings are also useful in the domain of emotion recognition since they can capture a large variety of behavioral clues that include facial expressions, eye movement, head movement and micro-expressions. Research in psychology has revealed that facial movements are closely linked to emotional conditions and facial expression is important in detecting emotional conditions.

Besides visual information, the audio signal connected to the video record type of information is also valuable emotionally. The production mechanisms of speech are usually modified by emotional states, which lead to modification of the pitch, tone, intensity, rhythm, spectral properties of voice. As an illustration, excitement can cause the voice pitch to elevate and the rate of speech to be more rapid whereas sadness can cause the voice energy to be less, and slower speech patterns.

Textual input, which could be optional, is also supported by the system and could be given in the process of inference. This textual data can be the conversation transcript or the comments left by users, or the description of the emotional state of the speaker. Even though this text is not utilized in training models, it may be useful in offering useful contextual information that enhances the interpretation of emotions in the real world. With the help of the combination of audiovisual cues and the contextual text, the system is capable of ensuring a deeper insight into emotional states.

3.2 Data Preprocessing

In order to have machine learning models process the collected data, the data will have to be pre-processed to be consistent and of quality. Raw audiovisual data can include noise, redundancy, or format variation that can impact negatively on model performance.

In the visual modality, the video input is first broken down into a series of images. The dimensions of the frames are resized and normalized to keep the input dimensions of neural network models constant. Normalization is used to stabilize training by making sure that the pixel values do not exceed an appropriate range. Frame sampling method can also be used to ensure that a constant temporal resolution is assumed with respect to various video samples.

The video recording is turned into an audio signal, which is then processed to produce a

form that can be consumed by deep learning models. The system does not directly analyze the raw waveform but instead converts the audio signal to a Mel spectrogram. A spectrogram is a representation of the frequency content of the signal as it varies with time and the Mel scale modifies the frequency representation to give a representation as perceived by humans.

Mel frequency scale can be calculated as follows:

$$\text{Mel}(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right) \quad (3.1)$$

where f is the original frequency in Hertz and $\text{Mel}(f)$ is the perceived pitch in the Mel scale. The model can target perceptually meaningful acoustic patterns which are relevant to emotional speech because of this transformation.

There is also pre-processing of textual inputs where they are used in the fusion stage where they are available. It is cleansed of irrelevant characters and normalized, and tokenized into smaller linguistic units, which can be processed by language models.

3.3 Deep Feature Extraction

The system then picks out the meaningful features of the audiovisual inputs with deep neural networks after preprocessing. Emotion recognition is an exception, and deep learning models tend to be more effective at it due to the ability to acquire hierarchical representations of complex patterns inherent in the raw data automatically.

3.3.1 Visual Feature Extraction

The visual arm of the system utilizes a three-dimensional convolutional neural network to take the frame sequences of the video. Whereas the traditional 2-dimensional convolutional networks operate on each image separately, the 3D convolutional networks can work on spatial and temporal dimensions. This enables the model to record dynamic facial expressions and motion patterns, which change with time.

The mathematical representation of a 3D convolution operation is given by:

$$V_{\text{out}}(x, y, t) = \sum_i \sum_j \sum_k W(i, j, k) \cdot V_{\text{in}}(x + i, y + j, t + k) \quad (3.2)$$

where V_{in} , W , and V_{out} denote the input video tensor, the convolution kernel, and the resultant feature map respectively. The indices i, j, k are spatial and temporal kernel dimensions. This operation allows the network to acquire spatiotemporal patterns of facial expression like smiling, frowning, or surprise.

The network acquires more and more abstract representations of facial behavior, through several layers of convolutional operations. The first few layers detect simple visual images including edges and textures whereas deeper layers detect more advanced facial movements and transitions in expression. The last result of the visual stream is a feature vector which represents a summary of the emotional data obtained out of the video sequence.

3.3.2 Audio Feature Extraction

The audio branch takes the Mel representation of a speech signal (in the form of spectrogram) and runs it through a convolutional neural network architecture. The spectrogram takes the audio waveform and converts it into a two-dimensional representation where one of the axes is the time and the other one is the frequency. The strength of every point indicates the amount of energy at a given frequency and time.

This representation is then analyzed by the convolutional network to identify trends associated with the speech features of emotion such as pitch, harmonic patterns, and energy distributions over time. In order to enhance learning stability and enable the network to learn richer feature representations, residual network archetypes are frequently deployed.

F_v and F_a are the feature vectors derived as a result of the visual and audio network respectively. These feature vectors can be represented as:

$$F_v = f_v(\text{Video}) \quad (3.3)$$

$$F_a = f_a(\text{Audio}) \quad (3.4)$$

where f_v and f_a are the feature extraction functions of the neural network models. These vectors are summaries on the emotional cues available in the visual and acoustic modalities.

3.4 Mechanism of Cross-Modal Attention

The system uses a cross-modal attention mechanism to be able to incorporate the features of the visual and audio features. This process enables the learning of the various modalities relationships dynamically in the model and also dictates the modality with relations that give more reliable information in a specific situation.

Attention mechanism can be mathematically modeled as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (3.5)$$

where Q , K , V , and d_k are respectively the query matrix, key matrix, value matrix and the dimensionality of the key vectors. In this process, the model will give more weight to those features which are more informative in recognition of emotions.

The visual features can be focused more, as an example, when the facial expression is visibly measured in video. Conversely, in the event of partial obstruction of the facial view, the system might be more dependent on the acoustic information available in speech.

3.5 Multimodal Feature Fusion

Once the relative values of each modality have been ascertained by the attention mechanism, the visual and audio feature vectors are fused to form a single multimodal representation. This process of fusion takes emotional cues of both modalities and combines them into a feature vector.

The fusion process may be denoted as:

$$F_{\text{fusion}} = W_v \cdot F_v + W_a \cdot F_a \quad (3.6)$$

where W_v and W_a are weighting parameters that determine the contribution of visual and audio features respectively. The obtained feature vector provides a complete representation of emotional cues that are extracted by audiovisual information.

3.6 Rule-Based Context Generation

Although deep neural networks offer strong feature representations, the decision-making process of such networks is not always interpretable. In order to overcome this shortcoming, the recommended methodology will use a rule-based context generation component generating human readable descriptions of emotional signals.

In the case of visual modality, facial landmark detection and blendshape estimation method are employed to quantify facial muscle movements about various expressions. These measures are evaluated through standardized rules which translate numerical measurements to descriptive words. As an example, eyebrows raised can be an indication of surprise, whereas tight lips can be an indication of anger.

The speech sound also is examined to derive the prosodic features of pitch, loudness, spectral energy, as well as speech intensity. These features are interpreted using rule-based thresholds to produce textual descriptions of the vocal behavior.

The descriptions that are created on the basis of the visual and audio messages are brought to a contextual summary that interprets the perceived emotional indicators. In case optional text provided by the user is given in the process of inference, then this text is added to this contextual summary to produce a complete contextual representation.

3.7 Multimodal Fusion With a Transformer Model

Transformer-based language model combines the results of the deep feature extraction stream and the rule-based context generation stream. Transformer architectures are especially efficient in multimodal fusion since they rely on self-attention to learn how to establish the relationship between diverse kinds of information.

Within the suggested scheme, the multimodal feature vector obtained based on the audio-visual data is projected into the embedding space of the transformer model. This estimated vector is inserted into the series of token embeddings that are passed through the transformer. The rule-based text analyzed context is tokenized and transformed into embeddings through the tokenizer of the model.

The transformer encoder then operates on this resulting sequence of embeddings and creates a contextual representation that combines both texture and text descriptions. This model is a representation of multifaceted linkages amid perceptual cues and semantics.

3.8 Emotion Classification

The resulting contextual representation of the transformer is then sent to a classification layer where the emotional category is to be predicted. This layer is usually a full-connected neural network with a softmax activation function.

The likelihood of both classes of emotions can be calculated as:

$$P(y_i | x) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (3.7)$$

where z_i is the score on the class i output and the denominator is the sum of the exponential scores of all the emotion classes. The given emotion is the one which is predicted based on the maximum probability of the class.

3.9 The Pros of the Proposed Methodology

The suggested methodology is beneficial in a number of ways over the conventional emotion recognition systems. Multimodality enhances the strength of emotion detection because it utilizes signals in numerous sources. Deep learning combined with rule-based reasoning leads to a better prediction accuracy and interpretable results. Moreover, the heterogeneous data types can be easily combined through the transformer-based fusion, which provides the possibility of interaction between the visual and audible signals and the contextual language data.

3.10 Summary of chapter

The given methodology is a multimodal emotion recognition framework being suggested that integrates deep neural feature extraction, rule-based contextual reasoning, and transformer-based fusion. The system can capture perceptual and semantic aspects of the emotional expressions by means of visual, acoustic, and textual information integrated in one architecture. The hybrid method allows better and context-specific emotion recognition and thus the system can be used in real-life scenarios where emotional cues can be intricate and multimodal.

Chapter 4

System Design

The general structure of the suggested multimodal emotion recognition framework. The proposed framework presents a hybrid multimodal architecture aimed at achieving resilient emotion recognition through the integration of deep neural representations, rule-based contextual reasoning, and transformer-based language model fusion. The architecture uses information from visual, acoustic, and textual sources to help the system capture both low-level perceptual cues and high-level contextual interpretations of emotional states.

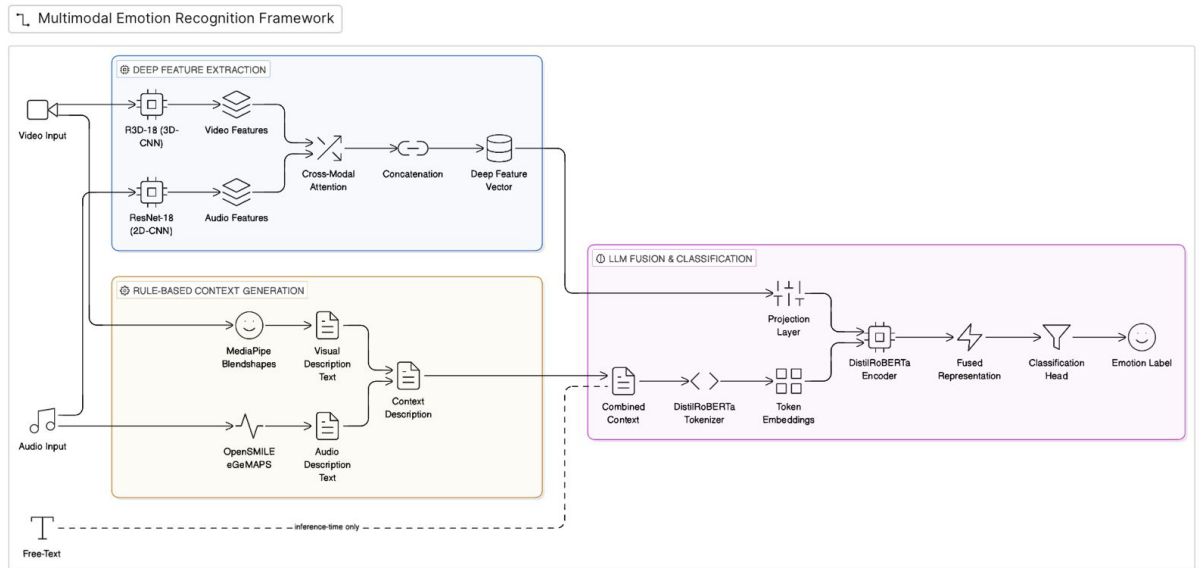


Figure 4.1: Architecture Diagram of the suggested multimodal emotion recognition framework

There are three main parts to the system:

- Multimodal Input Interface
- Two Streams for Parallel Processing
- Module for LLM-Based Fusion and Emotion Classification

This modular design lets the framework handle different types of inputs quickly and lets contextual information be added easily during inference.

4.1 Input Interface with Multiple Modes

The framework uses three different types of input that together give a wide range of emotional cues.

Input for Video

The video input is the main way to get visual emotional information. It records facial expressions, eye movements, head position, and micro-expressions that show how people feel. Facial muscle movements are closely linked to emotional expressions, according to well-known psychological models like the Facial Action Coding System (FACS). This makes emotional cues in the visual domain very useful.

The video input is usually shown as a series of frames taken from the original recording. These frames keep spatiotemporal information, which is important for recognizing facial expressions that change, like smiling, frowning, or being surprised.

Input for Audio

The audio modality gives extra emotional cues based on the prosody and vocal characteristics of speech. Emotional states affect the mechanisms that produce voice, which changes the pitch, tone, loudness, rhythm, and spectral properties.

The system can find patterns in the acoustic signal that are linked to emotional expression, such as:

- High pitch when you're excited
- Less energy when you're sad
- Irregular prosody when angry or stressed

This modality thus offers an additional avenue for emotion inference, especially beneficial when visual cues are unclear or partially obstructed.

Free-Text Input (Only at Inference Time)

The architecture also supports an optional Free-Text Input that can be used at inference time in addition to audiovisual inputs. This input could have comments from users, transcripts of conversations, or notes that give semantic clues about how the speaker is feeling.

It is important to note that the free-text input is not used to train the model. Instead, it is only added during inference to help the model understand the context better. The dashed border around this input in the architecture diagram makes it clear that it is only for inference.

The system can adjust to real-world situations where external contextual information may be available by adding optional textual context.

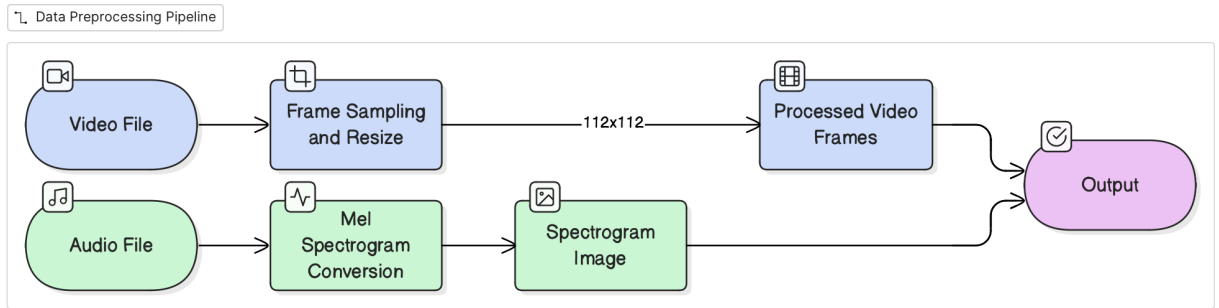


Figure 4.2: Data Preprocessing

4.1.1 Streams for Parallel Processing

The architecture uses two parallel processing streams to handle the multimodal inputs. Each stream is made to capture a different type of emotional information.

The first stream is about getting deep neural features, and the second stream is about making rules-based context that can be understood.

4.2 Stream A: Getting Deep Features

Stream A uses deep neural networks to learn high-level multimodal representations directly from audiovisual data. This stream captures complicated emotional patterns that rule-based systems might not be able to describe easily.

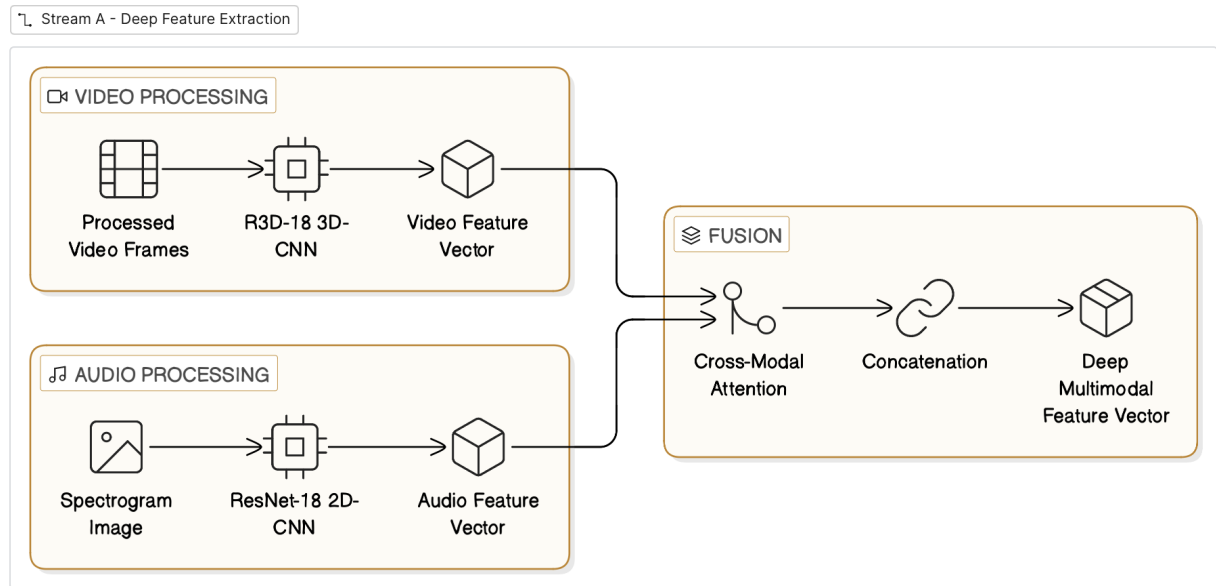


Figure 4.3: Stream A - Deep Feature Extraction

Using R3D-18 to Get Visual Features

The R3D-18 architecture, a 3D Convolutional Neural Network (3D-CNN) made for learning spatiotemporal features, processes the video input.

3D CNNs do convolutions across both spatial and temporal dimensions, which is different from traditional 2D CNNs that only work on single frames. This lets the model capture motion dynamics and temporal dependencies in the video sequence.

The network learns representations that encode through a hierarchy of convolutional layers:

- Movements of the muscles in the face
- Changes in expression
- Intensity of motion
- Patterns of facial activity over time

The last thing that the visual branch does is make a video feature vector that shows the learned emotional traits taken from the input video.

Using ResNet-18 to Get Audio Features

First, the audio signal is changed into a spectrogram, which turns the time-based waveform into a picture of frequency energy over time.

A ResNet-18 architecture, which is a deep residual convolutional network that is often used for visual pattern recognition tasks, then processes this spectrogram image. ResNet's residual connections help the network learn deeper representations and stop the gradient from disappearing.

The ResNet-18 network gets acoustic feature embeddings that show patterns like:

- Harmonic structures
- Distribution of spectral energy
- Change in pitch
- Changes in intensity over time

This branch's output is an audio feature vector that shows the emotional cues in the speech signal.

Cross-Modal Attention System

After the visual and audio feature vectors are taken out, they go through a cross-modal attention module. This mechanism enables the model to dynamically learn relationships between the two modalities.

The attention mechanism helps the system figure out which modality gives more accurate information in a given situation. For instance:

- When facial expressions are clearly visible, visual cues may receive higher attention.
- When visual signals are unclear, acoustic prosody might give you more information.

The model learns to adaptively weight features from both modalities through this process, which improves the quality of the final representation.

Fusion of multimodal features

After cross-modal attention, the features that were focused on from the audio and video branches are combined to make a deep multimodal feature vector.

This vector is a high-dimensional embedding that holds the combined emotional information taken from both visual and acoustic sources.

After that, this feature vector is sent to the fusion module, where it will be combined with contextual text information.

4.3 Stream B Rule-Based Context Generation

While Stream A focuses on deep representation learning, Stream B generates clear contextual descriptions based on specific rules applied to extracted audiovisual features. This stream provides human-readable summaries of emotional signals, allowing the model to include symbolic contextual information.

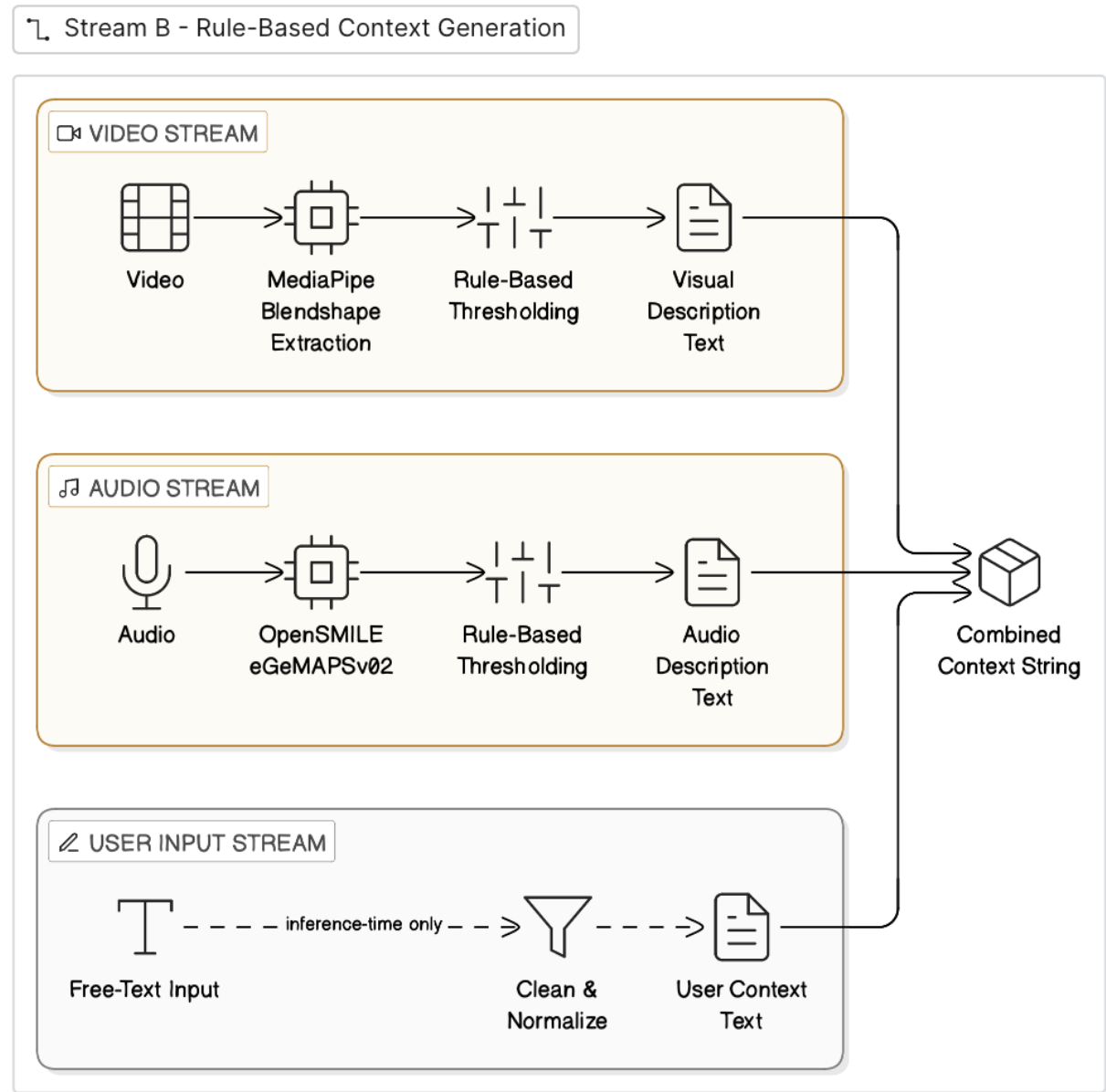


Figure 4.4: Stream B - Rule-Based Context Generation

Visual Context Generation Using MediaPipe Blendshapes

The video input is processed through MediaPipe Blendshape extraction, which estimates facial muscle activations linked to various facial expressions. Blendshapes measure the movement of

facial parts like:

- Eyebrow raises
- Eye widening
- Mouth stretching
- Lip compression

These numerical values are assessed using rule-based thresholds to convert facial activity into descriptive text statements. For example:

- Raised eyebrows may suggest surprise
- Lip tightening may suggest anger
- Relaxed facial muscles may suggest neutral or calm states

This stage produces a visual description text summarizing the observed facial expressions.

Audio Context Generation Using OpenSMILE eGeMAPS

The audio stream is analyzed with OpenSMILE, a common toolkit for extracting acoustic features related to emotion recognition. Specifically, the system extracts the eGeMAPS (extended Geneva Minimalistic Acoustic Parameter Set), which includes features known to connect with emotional expression.

These features include:

- Fundamental frequency (pitch)
- Loudness
- Spectral flux
- Formant frequencies
- Voice quality parameters

Rule-based thresholding is then applied to interpret these values and create text descriptions of vocal behavior, such as:

- High pitch variability

- Increased vocal energy
- Reduced speaking intensity

The resulting audio description text captures acoustic emotional signals.

Context String Construction

The visual and audio descriptions from this stream are combined into a single context description string. This string provides a summary of detected emotional signals taken from the audiovisual inputs. If present, any Free-Text Input from the user during inference is added to this context string, resulting in a Combined Context String with both system-generated and user-provided contextual information.

4.4 LLM-Based Fusion and Emotion Classification

The final stage of the architecture integrates the outputs of both streams using a modified transformer-based language model known as DistilRoBERTa.

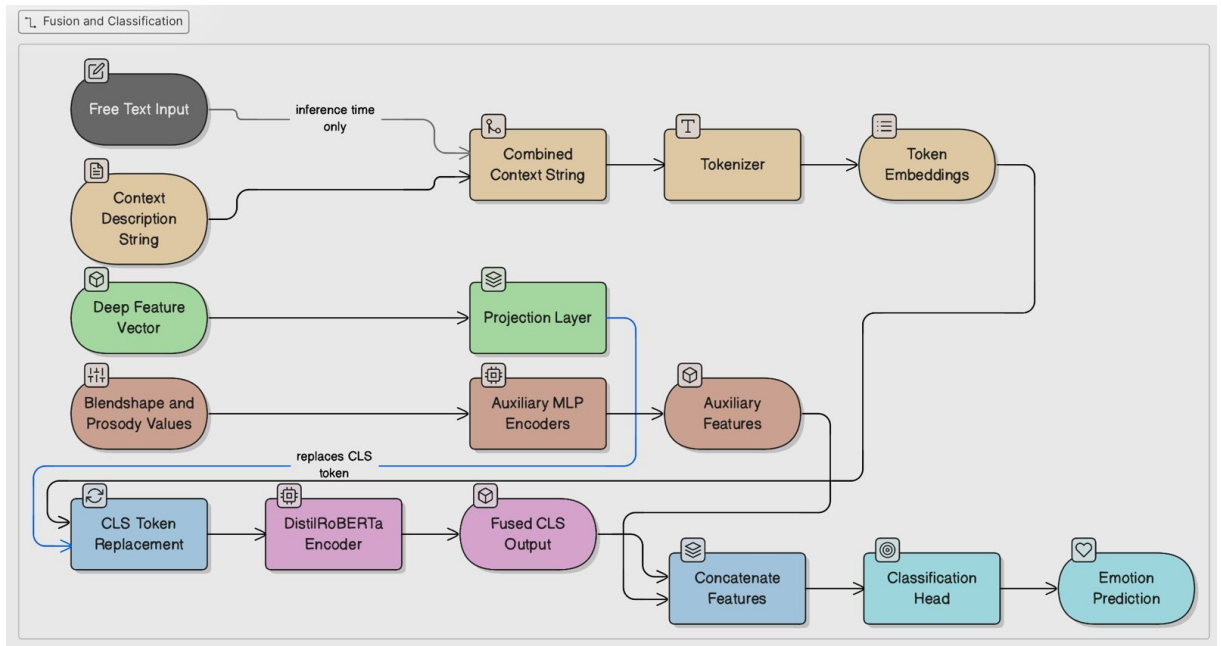


Figure 4.5: Fusion and Classification Module

Multimodal Feature Projection

The deep multimodal feature vector created by Stream A is passed through a projection layer that maps it into the embedding space of the DistilRoBERTa model. This projected vector

replaces the standard [CLS] token embedding in the token sequence. The [CLS] token in transformer models normally acts as a global representation used for classification tasks.

By swapping this token with the projected multimodal feature vector, the architecture allows the transformer to directly use audiovisual information in its contextual representation.

Text Tokenization and Embedding

The Combined Context String generated in Stream B is processed using the DistilRoBERTa tokenizer, which changes the text into a sequence of tokens. Each token is turned into a token embedding, creating a sequence of contextual representations that show the emotional signals detected from the audiovisual analysis and optional user-provided text.

Transformer-Based Multimodal Fusion

The modified token sequence, which includes:

- The projected multimodal feature token
- Contextual text token embeddings

is processed by the DistilRoBERTa encoder. Through multiple transformer layers, the encoder performs self-attention operations, allowing interactions between textual context and audiovisual feature representations. This process results in a combined contextual representation that integrates both modalities.

Emotion Classification

The final fused representation is sent to a classification head, usually arranged as a fully connected neural network layer followed by a softmax activation function. This part maps the fused representation to a set of defined emotion categories, producing the final emotion label prediction.

4.5 Sequence Diagram Description

The sequence diagram illustrates the step-by-step interaction between different components of the multimodal emotion recognition system. It captures the flow of data from the user input stage to the final emotion prediction.

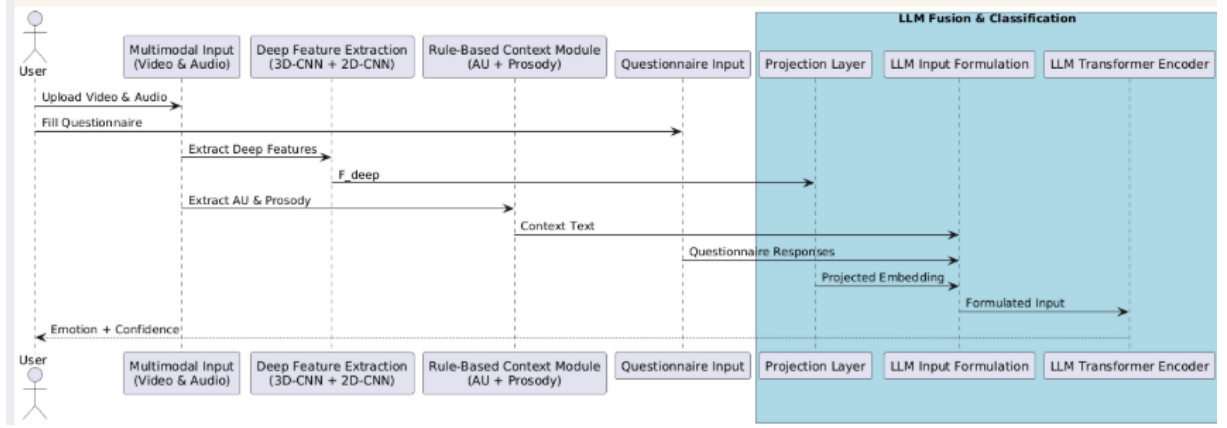


Figure 4.6: Sequence Diagram of Multimodal Emotion Recognition System

4.5.1 Step-by-Step Workflow

1. **User Input:** The process begins when the user uploads video and audio data into the system. Simultaneously, the user fills out a questionnaire to provide additional contextual information.
2. **Multimodal Input Processing:** The uploaded video and audio are forwarded to the multimodal input module, which prepares the data for further processing.
3. **Deep Feature Extraction:** The system extracts deep features from the input data:
 - Visual features are extracted using 3D-CNN and 2D-CNN models.
 - Audio features are extracted from speech signals.

These are represented as deep feature vectors (F_{deep}).

4. **Rule-Based Context Generation:** In parallel, the system processes facial Action Units (AU) and audio prosodic features such as pitch and tone. This information is converted into contextual text using rule-based techniques.
5. **Questionnaire Processing:** The responses from the questionnaire are collected and formatted as structured input data.
6. **Data Flow to Fusion Layer:** The extracted deep features, contextual text, and questionnaire responses are forwarded to the fusion module.
7. **Projection Layer:** All inputs are transformed into a unified embedding space. This ensures compatibility between heterogeneous data types.

8. **LLM Input Formulation:** The projected embeddings and contextual information are combined to form a structured input suitable for the Large Language Model.
9. **LLM Transformer Encoder:** The structured input is processed by a transformer-based encoder, which learns relationships across modalities and context.
10. **Emotion Prediction Output:** Finally, the system outputs the detected emotion along with a confidence score, which is presented to the user.

4.5.2 Software and Hardware Requirements

Software Requirements

The development and implementation of the proposed multimodal emotion recognition system require the following software components:

- **Operating System:** Windows, Ubuntu, or macOS
- **Programming Language:** Python 3.9 or higher
- **Frameworks:** PyTorch or TensorFlow for deep learning model development
- **Libraries:** OpenCV (for image processing), Librosa (for audio processing), NumPy and Pandas (for data handling)
- **LLM Support:** HuggingFace Transformers for integrating large language models
- **Development Tools:** Visual Studio Code or Kaggle Notebook
- **GPU Support:** CUDA Toolkit for accelerating deep learning computations

Hardware Requirements

The system requires the following hardware configuration for efficient performance:

- **Processor:** Intel i7 or AMD Ryzen 7 (minimum Intel i5)
- **RAM:** 16 GB recommended (minimum 8 GB)
- **GPU:** NVIDIA RTX 3060 or higher with at least 8 GB VRAM
- **Storage:** 512 GB SSD
- **Input Devices:** Webcam and Microphone for capturing video and audio inputs

4.6 Key Contributions of the Architecture

The proposed architecture introduces several new design elements that enhance multimodal emotion recognition performance:

- Hybrid representation learning that combines deep neural features and rule-based contextual descriptions.
- Cross-modal attention that enables the integration of audio and visual signals.
- Transformer-based multimodal fusion using a modified DistilRoBERTa architecture.
- Direct multimodal feature injection by replacing the [CLS] token embedding.
- Contextual augmentation at inference time through optional free-text input.
- Clear context generation using rule-based audiovisual descriptions.

4.7 Work Breakdown Structure

The project was systematically divided among team members to ensure efficient development and timely completion. Each member was assigned specific responsibilities based on different modules of the system.

4.7.1 Team Roles and Responsibilities

- **Savio Tomy:** Responsible for deep visual feature extraction using 3D-CNN and 2D-CNN models, including preprocessing of video data and model implementation.
- **Volwin Shaji:** Handled the semantic stream, including visual feature pipeline design and integration of extracted features into the system.
- **Yuhan John:** Focused on audio feature extraction, including processing speech signals using Librosa and generating deep audio representations.
- **V Karthik:** Responsible for system integration, LLM-based fusion, questionnaire processing, and final emotion classification using transformer models.

4.7.2 Task Distribution

The project was executed in multiple phases:

- **Foundation and Scoping:** Literature review, problem formulation, and dataset collection.
- **Semantic Stream:** Visual and audio feature extraction pipelines along with rule-based text conversion.
- **Deep Feature Stream:** Implementation of deep learning models for extracting high-level features.
- **Final Fusion:** Integration of multimodal features and classification using LLM.

4.8 Project Timeline

The project development was carried out over two semesters. The initial stage focused on requirement analysis, literature review and system design. In the Later Stage the Intergration of Modules, evaluation and preparation of project documentation.



Figure 4.7: Project Timeline Showing Task Distribution and Phases

4.9 Summary of Chapter

The overall architecture presents a well-rounded multimodal framework that combines deep learning, symbolic reasoning, and transformer-based language modeling to achieve strong emotion recognition. By merging audiovisual signals, contextual descriptions, and optional text input, the system can capture both low-level perceptual features and high-level semantic context. This results in better emotional understanding and prediction accuracy.

Chapter 5

Results and Discussion

In this part, the experimental findings of the suggested multimodal emotion study framework are provided. This system combines visual features that are obtained by analyzing video frames, acoustic features that are obtained by analyzing speech signals and contextual information, which is obtained by rule-based analysis. The experiments show that the model can learn significant emotional representations based on audiovisual data and make accurate prediction of emotions.

5.1 Model Training Performance

The RAVDESS speech and song data of emotion were used to train the model. The dataset is represented by several actors with the manifestation of various emotional states in audio-visual records. To train and validate, the dataset was separated by actor to prevent the leakage of identity between training and validation sets.

Training was carried out on actors 1–20 and validation was carried out on actors 21–24. This division makes sure that the model is tested on people that have not been seen before.

It was a training process that entailed the extraction of multimodal features of both video and audio streams, multimodal fusion and classification.

The multimodal feature vector is a combination of audio and visual data that is 1024-dimensional. The model gives predictions in eight categories of emotions.

The system took 2036 samples of training and 416 samples of validation during training. The overall accuracy of the training was about 95.38 percent whereas the validation accuracy was 71.63 percent. The disparity between training and validation accuracy means that the model has learned the emotional patterns correctly but it can also be further refined with new regularization or more training data.

```
Combined Feature Vector Shape: torch.Size([4, 1024])
Prediction Output Shape: torch.Size([4, 8])
Training on Actors: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
Validating on Actors: [21, 22, 23, 24]
Training on 2036 samples, Validating on 416 samples.
Starting training on cuda...

Results: Train Loss: 0.1717, Train Acc: 95.38% | Val Loss: 0.9362, Val Acc: 71.63%
-----
✅ Model Saved (New Best Validation Loss)
```

Figure 5.1: Model training output showing combined feature vector

5.2 Emotion Prediction Example

To work out the functioning of the proposed framework, an example of a prediction output is provided. The video frames are compared with the speech signal to give an emotion prediction using the system.

The former is represented by the first component of the output and shows the log-Mel spectrogram of the incoming audio signal, the frequency distribution of the speech over time. This form of representation of spectrogram helps the neural network to identify the changes in pitch, tone and speech energy, when in a particular emotional state.

The other output section displays the video frames obtained out of the input sequence. These frames are used to capture the facial expressions and movements which give visual signals of the emotional states.

The fused multimodal feature vector is a high-dimensional representation of both the visual and acoustic features. The resultant combination representation enables the system to examine facial expression and speech attributes relationships.

The last step gives a probability distribution of the emotion categories. In the case provided, the system was able to label the emotion of Sad with a confidence score of about 78.3, which is equivalent to the ground truth.

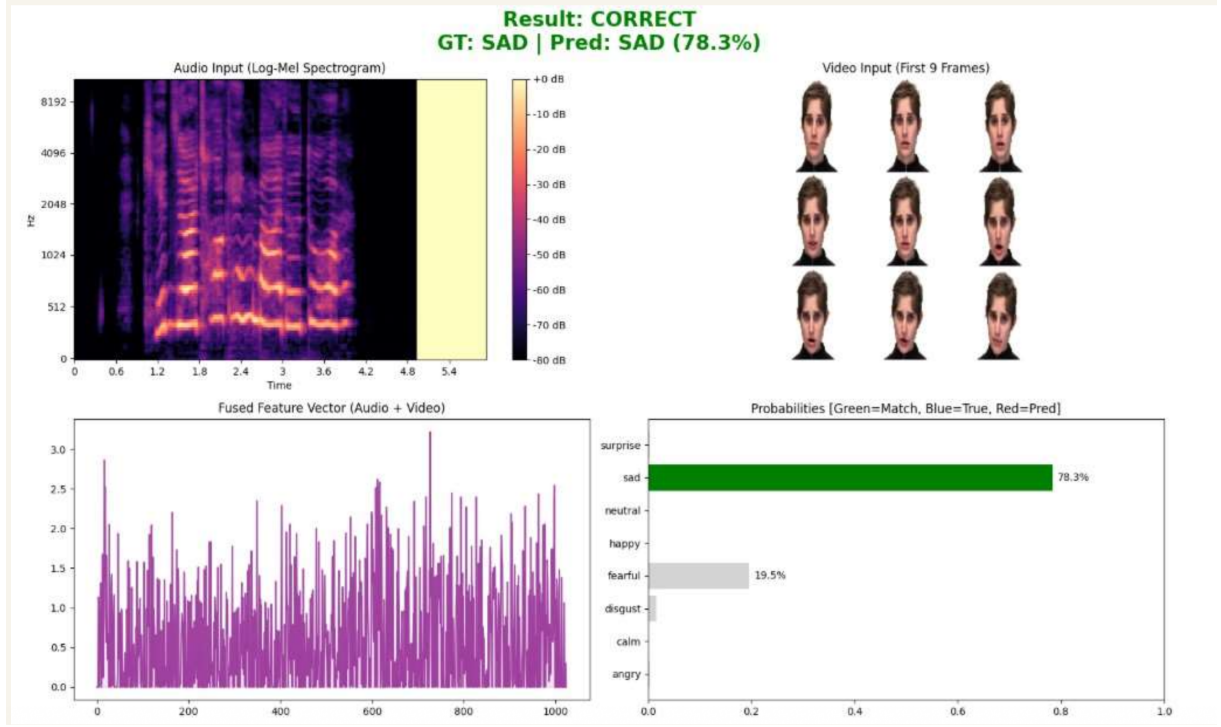


Figure 5.2: Emotion prediction output

5.3 Audio Feature Analysis

The system also does elaborate extraction of acoustic features to decode the emotional speech vocal characteristics. The OpenSmiLE toolkit is used to extract these features, and the eGeMAPS parameter set, typical to speech emotion recognition studies, is typically used.

Some of the acoustic cues obtained out of the speech signal are equal sound level, pitch variation, spectral flux and Mel-frequency cepstral coefficients (MFCCs). The characteristics of the speech signal that are described by these features include the loudness, the stability of the pitch, and the spectral structure.

The system transforms these numerical characteristics into textual descriptions in human readable format based on pre-defined thresholds and text rule-based analysis. In this instance, the description generated would be something with moderate volume and comparatively steady pitch.

These textual signals will be in turn added to the contextual prompt that the transformer-based fusion module will be employing to enable the system to read emotional indicators in a more significant manner.

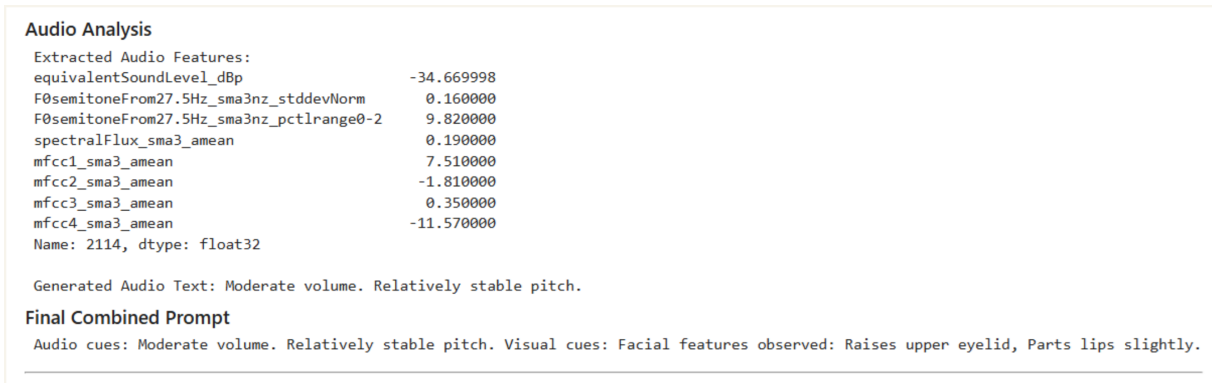


Figure 5.3: Audio analysis: eGeMAPS features and multimodal prompt

5.4 Visual Feature Analysis

Besides the acoustic analysis, the system also analyzes facial expressions with respect to facial action units recognized with the MediaPipe-based blendshape estimation. Facial action units are definite facial muscle expressions, and they are extensively applied in the Facial Action Coding System (FACS).

The system identifies different facial gestures that include eyebrow raising, tightening of lips, cheek movement and mouth opening. The individual action units detected are matched with a value of intensity which shows the strength of the activation of the facial muscle.

With these values, the system creates a text summary of the facial behavior of the subject. In the given result, the description that is generated refers to the subject lifting the upper eyelid and slightly opening the lips.

These facial expressions give further information to the decoding of the emotional state of the person and supplement the acoustic features derived out of the speech signal.

5.5 Context Multimodal Generation

The textual descriptions that are produced after audio and visual analysis are put together to produce a contextual prompt. This summarizes the emotional cues that are captured in the two modalities.

Contextual description example may consist of the following statements:

Audio stimuli are evident of moderate volume with relatively constant pitch. There are visual clues of elevated eyelid in the upper part and a small separation of the lips.



Figure 5.4: Visual analysis for Angry sample: facial AUs and description

This is a combined context in which it is sent to the transformer-based fusion module. This contextual information is processed by the language model along with deep neural features to enable the system to combine quantitative and qualitative emotional information.

5.6 Final Result

The system developed offers the user an interactive platform to carry out multimodal emotion recognition with respect to video, audio, and contextual analysis. The interface combines several elements that represent the processing pipeline and display the estimated emotional state and the confidence scores as well. This interface will be aimed at enabling users to see the way that the system processes audiovisual inputs and the way it generates emotion predictions in real time. The primary part of the interface shows the input video upon which emotions analysis is performed. Facial expression and speech signs of an object are present in the video and are the main agents of emotional data. In processing, the system uses frames that are extracted in the video to analyze the facial expression even as it extracts the audio signal to analyze the acoustic features. These characteristics are summed up by the trained multimodal model to come up with an emotion prediction. The analysis of the video clip presented in the interface in the example is based on the facial cues of sadness displayed by the person in it. The model works on the visual characteristics of the facial expression and integrates it with the acoustic characteristics based on the speech signal. The system forecasts the emotional condition of the subject after multimodal fusion. The emotion predicted is presented on the video frame itself and also accompanied by a confidence score. In the given case, the system recognizes the emotion as Sad and the confidence level is 79.4. The outcome of this result implies that

the observed model shows a strong relationship between the discerned facial expressions and speech traits and the patterns that are normally present in sad emotional states. In addition to the video display, the interface gives a panel where the probability distribution of all possible emotion classes can be seen. These are: Angry, Calm, Disgust, Fearful, Happy, Neutral, Sad and Surprise. A probability value is given to each emotion and it is an expression of the probability of that emotional state according to the features extracted. The probability bars enable the users to see the degree to which the model is taking into account every emotion before making the eventual prediction. According to the probability distribution in the case, Sad emotion is most likely to be found than any other emotion, including Angry, Disgust, and Surprise, which are rated much lower. This distribution proves that the model is able to differentiate between the various emotional patterns and choose the most appropriate class. The other element of the interface is the activity log that captures the internal processing steps that have been undertaken by the system. Activity log enhances the visibility of the activity since it displays the operations of the activity, which include video pick, audio extraction, feature processing, and prediction generation. The log enables users to comprehend the sequential process of the operations done during the recognition of emotions. When the model has been loaded successfully and is ready to be used in prediction, this is also shown in the interface. After selecting a video, the system automatically preprocesses, extracts features and fuses multimodal features, then it provides the ultimate prediction. The processing pipeline is also highly efficient such that it can be used to generate predictions in a short period of time. In general, the system output in the end shows that the suggested multimodal emotion recognition framework can handle audiovisual data, and it can make valid predictions of emotions. The system can measure a variety of elements of the emotional behavior of a human being through the combination of facial expression analysis and speech-based emotional cues. The probability scores and activity logs visualization also improves the interpretability of the system, as it will enable the user to have a more intuitive grasp of the way in which the prediction is obtained. This finding supports the idea that resorting to the integration of various modalities enhances the efficiency of the system to identify the state of emotion as opposed to utilizing one source of information. The interface thus has the purpose of a demo, as well as a functional use of the multimodal emotion recognition technology.

5.7 Discussion

The results of the conducted experiment indicate that the proposed hybrid architecture can effectively combine deep neural feature extraction and rule-based contextual reasoning. The

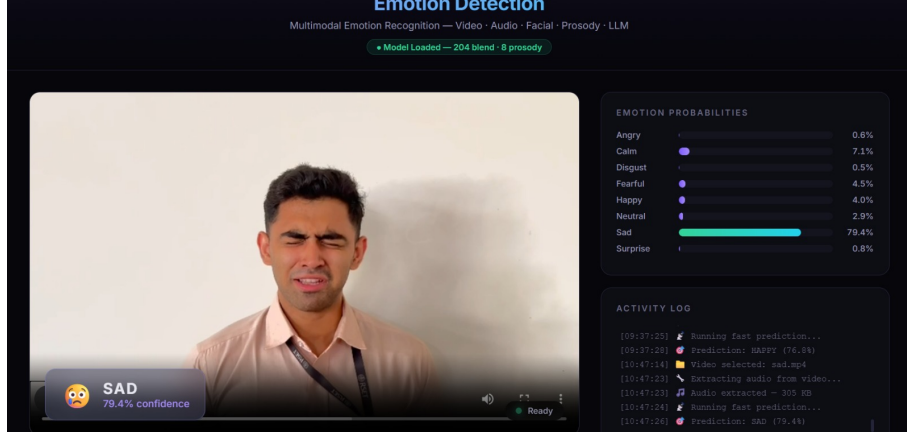


Figure 5.5: Final Emotion Prediction

model can observe emotional indicators in the facial expressions, voice features, and textual description of the context.

The multimodal fusion process will enable the system to extract the relationship between the various modalities and give credible predictions of emotions. Even though the accuracy of the training is relatively high, the validation accuracy is lower, which shows that it can be further refined by means of further data augmentation, better regularization methods, or training with bigger and more diversified datasets.

Comprehensively, the findings are validated to show that the consistency and readability of the emotion recognition systems can be enhanced significantly when audiovisual cues are combined with the interpretation of context.

5.8 Summary of Chapter

The experimental evaluation of the proposed multimodal emotion recognition framework demonstrates that integrating visual, acoustic, and contextual modalities significantly enhances the accuracy and interpretability of emotion prediction. The system successfully learned meaningful emotional representations from audiovisual data, with the multimodal fusion approach enabling the model to capture complementary cues from both facial expressions and speech characteristics. The rule-based contextual stream further enriched the predictions by generating human-readable descriptions of detected emotional signals, which were effectively combined with deep neural features through the transformer-based fusion module. Overall, the results confirm that the hybrid architecture produces robust and reliable emotion predictions, and highlight the clear advantage of multimodal integration over single-modality approaches.

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

The proposed system presents an efficient framework for enabling seamless handover and energy optimization in UAV-assisted communication networks. By integrating a multi-criteria decision-making approach with altitude-aware relay selection and adaptive base station association, the system ensures improved connectivity, reduced handover failures, and optimized energy consumption. The two-layered air-ground architecture further enhances coordination between UAVs and ground infrastructure, allowing effective load distribution and real-time communication support. Simulation results demonstrate that the system is capable of maintaining stable communication links while minimizing unnecessary handovers and conserving UAV energy, thereby improving overall network performance.

6.2 Key Contributions

The key contributions of this work include the design of a novel UAV-assisted network architecture that supports efficient handover and relay-based communication, the implementation of a multi-criteria decision engine for intelligent base station selection, and the development of an energy-aware mechanism that enhances UAV operational lifetime. Additionally, the system introduces altitude-aware relay selection, which improves communication reliability and reduces signal degradation. The integration of priority-based decision-making further ensures that critical communication requirements are handled effectively.

6.3 Impact of the Proposed System

The proposed system has significant potential in real-world applications such as disaster management, emergency communication, surveillance, and remote area connectivity. By improving handover efficiency and reducing energy consumption, the system contributes to more reliable and sustainable UAV network operations. Its ability to adapt to dynamic network conditions

makes it suitable for next-generation wireless communication systems, including integration with advanced technologies like 5G and beyond.

6.4 Limitations

Despite its advantages, the proposed system has certain limitations. The evaluation is primarily based on simulation, which may not fully capture real-world environmental conditions such as weather variations, physical obstacles, and hardware constraints. The computational complexity of the decision-making process may increase with large-scale deployments, potentially affecting real-time performance. Additionally, security aspects such as protection against cyber-attacks and unauthorized access have not been fully addressed in the current implementation.

6.5 Future Scope

Although the proposed system demonstrates effective handover management and energy-efficient communication in UAV-assisted networks, there remains significant scope for further research and enhancement to improve its practicality, intelligence, and scalability. Future work can focus on integrating advanced machine learning and deep learning techniques, such as reinforcement learning, federated learning, and predictive analytics, to enable real-time adaptive decision-making for handover management, relay selection, and resource allocation based on continuously changing network conditions, UAV mobility, and user demand patterns. The system can also be extended to support large-scale deployments involving a dense population of UAVs, requiring the development of distributed and low-complexity algorithms to ensure efficient coordination, reduced latency, and minimal communication overhead in highly dynamic environments. Furthermore, transitioning from simulation-based evaluation to real-world implementation using physical UAVs, sensors, and ground stations will provide valuable insights into practical challenges such as signal interference, weather conditions, hardware constraints, battery limitations, and regulatory compliance.

In addition, future enhancements can focus on strengthening the security and privacy aspects of the system by incorporating robust encryption schemes, blockchain-based secure communication frameworks, intrusion detection systems, and authentication protocols to safeguard against cyber-attacks, spoofing, and unauthorized access. Energy efficiency can be further improved by integrating energy harvesting mechanisms such as solar-powered UAVs, as well as developing intelligent energy-aware scheduling and routing techniques that dynamically balance performance and power consumption. The integration of the proposed system with emerging

communication technologies such as 5G and beyond (6G) networks presents another promising direction, enabling the use of advanced features like network slicing, edge computing, ultra-reliable low-latency communication (URLLC), and AI-driven network orchestration to enhance system performance and reliability.

Moreover, future work can explore advanced UAV mobility and trajectory optimization techniques that consider factors such as user distribution, terrain, obstacles, and traffic demand to maximize coverage and minimize energy consumption. The incorporation of cooperative multi-UAV communication strategies and swarm intelligence can further enhance system robustness, fault tolerance, and load balancing capabilities. Another potential area of research includes the development of context-aware and priority-based service models that can dynamically adapt to different application requirements, such as emergency communication, disaster management, and real-time surveillance. Finally, extending the system to support interoperability with other heterogeneous networks, including IoT devices and satellite communication systems, can significantly broaden its applicability and effectiveness in next-generation wireless communication scenarios, ultimately making the proposed model more scalable, resilient, and suitable for real-world deployment.

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Appendix A: Presentation

MULTIMODAL EMOTION RECOGNITION USING AUDIO, VIDEO AND TEXT FUSION

FINAL PRESENTATION

Guide:
Dr. Preetha K G

Team Members:
Savio Tomy
V Karthik
Volwin Shaji
Yuhan John

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INTRODUCTION

- Emotion recognition is an essential component of human-computer interaction, enabling machines to understand and respond to human feelings.
- Traditional systems relying on a single modality (audio, video, or text) often fail to capture the full emotional context, leading to inaccuracies.
- Our project focuses on multimodal emotion recognition, integrating audio, visual, and textual data to improve accuracy and reliability.
- By combining multiple sources of information, the system aims to mimic human perception more closely and enable applications in healthcare, education, entertainment, and customer support.

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PROBLEM DEFINITION

- Current emotion recognition systems are predominantly unimodal, relying on either facial expressions, speech, or text, which limits their effectiveness.
- These systems struggle to interpret emotions accurately in real-world scenarios, where context, tone, and subtle cues matter.
- Objective: Design a robust multimodal emotion recognition system that fuses audio, visual, and textual features to provide more accurate and context-aware emotion detection.
- Applications include mental health monitoring, emotion-aware AI in education, enhanced human-computer interaction, and entertainment analytics.

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IMPLEMENTATION

SCOPE

- Accepts video input containing synchronized audio and speech for emotion analysis
- Performs preprocessing of audio, video frames, and text transcripts to ensure data consistency
- Extracts emotional features using 3D-CNN for video and 2D-CNN for audio
- Generates rule-based contextual descriptions using facial action units and audio prosody
- Applies LLM-based multimodal fusion to combine deep features and contextual information
- Classifies emotions with confidence scores
- Evaluates system performance using benchmark datasets such as RAVDESS

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LITERATURE SURVEY

Paper Title	Methodology	Strengths	Limitations
Multimodal Attention-Mechanism for Temporal Emotion Recognition[1]	Attention-based fusion of speech + facial data across time sequence	- Attention highlights important modalities/timesteps - Better temporal modeling - Improves recognition accuracy vs. simple fusion	Training requires large labeled data - Risk of overfitting - High computational overhead
Multi-Modal Emotion Recognition From Speech and Facial Expression Based on Deep Learning[2]	Dual CNN (for speech spectrogram & facial features) + joint feature fusion	Exploits complementary cues from speech & face - End-to-end deep learning - Higher accuracy on benchmark datasets	Sensitive to noise (audio/video) - Requires balanced multimodal datasets - Computationally expensive

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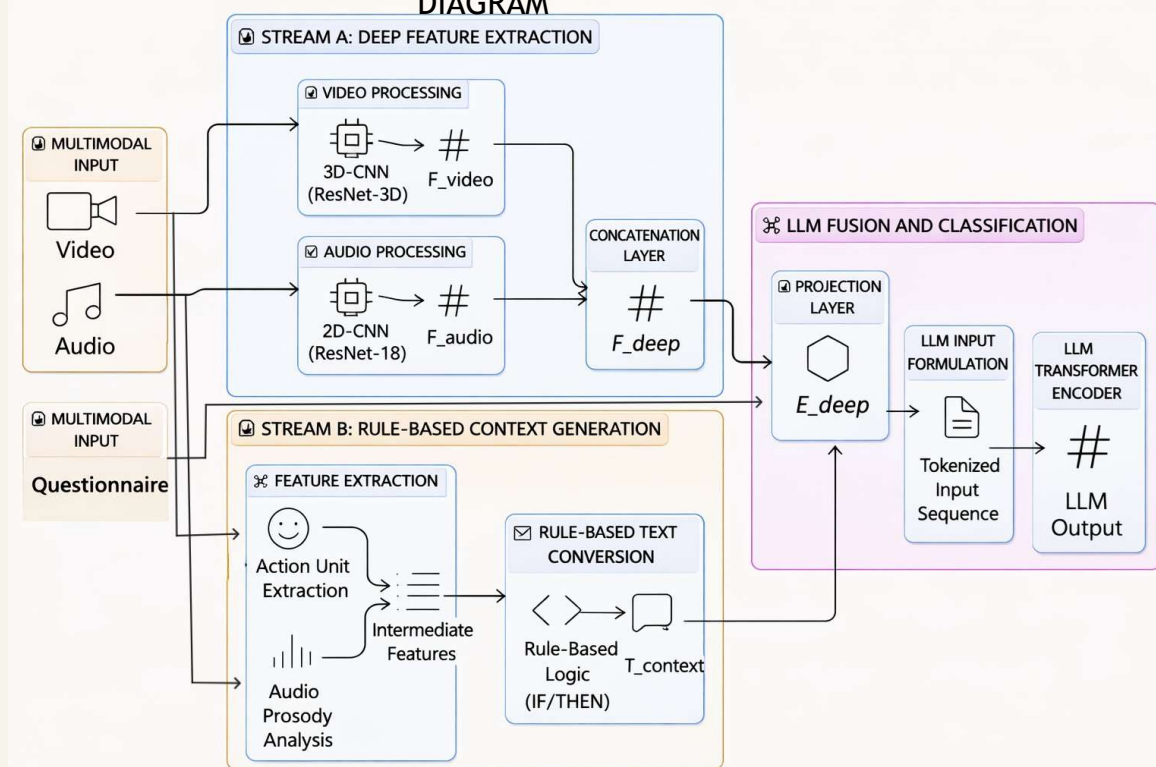
Paper Title	Methodology	Strengths	Limitations
Multimodal Deep Learning for Emotion Recognition[3]	Speech → 1D CNN + BiLSTM; Facial expressions → small-kernel 2D CNN; Feature-level fusion via DNN	- Effective multimodal fusion (70.24% WA on IEMOCAP) - Captures temporal + spatial cues - Robust over unimodal	- High computational cost - Dataset-specific tuning - Limited interpretability
Integrating Deep & Shallow Models for Multi-Modal Depression Analysis (Hybrid Architectures) [4]	Integrating Deep & Shallow Models for Multi-Modal Depression Analysis (Hybrid Architectures)	Combines behavioral + linguistic cues - Novel HDR + PV features - Better accuracy (F1 = 0.746 on AVEC2016) - Scalable hybrid design	Dataset-specific (AVEC2016) - Resource heavy - Relies on transcript quality

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Paper Title	Methodology	Strengths	Limitations
Multimodal Emotion Feature Fusion: An LLM-Based Approach[5]	Uses rule-based systems to convert non-verbal cues into text. - Extracts audio features (loudness, spectral flux, MFCCs, pitch stability, emphasis) and visual features (facial action units) from video. - Combines these features into a textual	• High accuracy – 93.18% (RAVDESS) and 93.69% (BAUM-1), comparable to state-of-the-art. • Unified fusion – converts audio & visual cues into text for LLM-based modeling. • Scalable – rule-based textualization allows easy extension to other modalities.	Signal loss – text conversion may drop subtle audio/visual details. Fusion limits – treating all data as text may weaken modality-specific strengths.

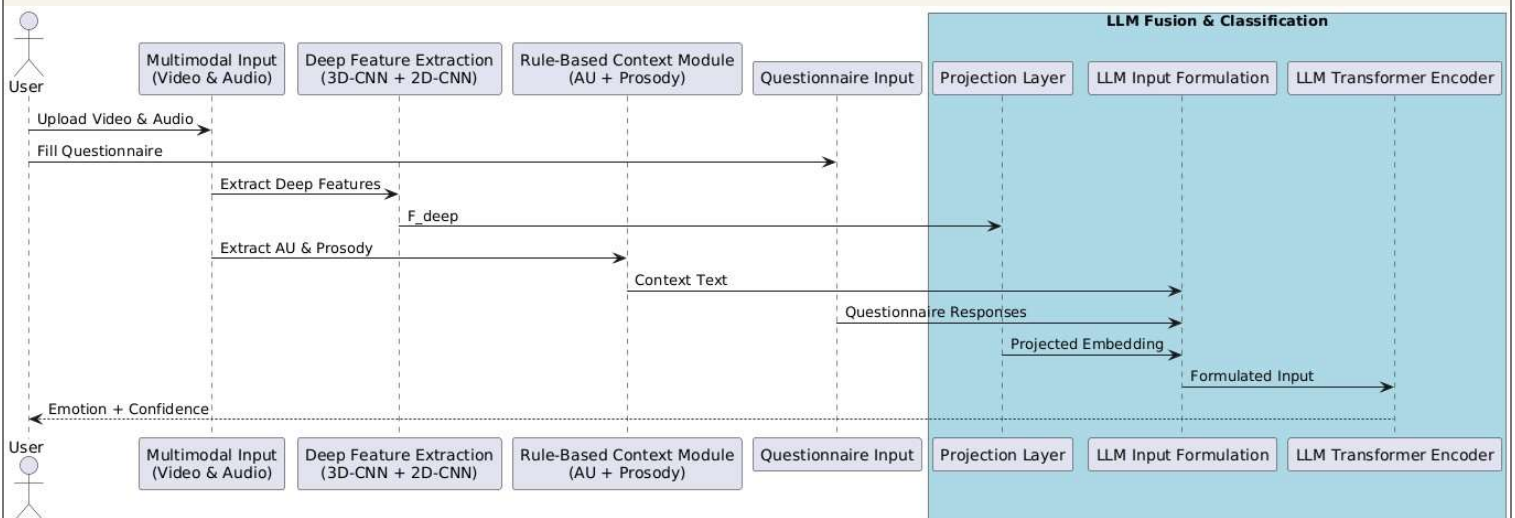
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ARCHITECTURE DIAGRAM



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SEQUENCE DIAGRAM

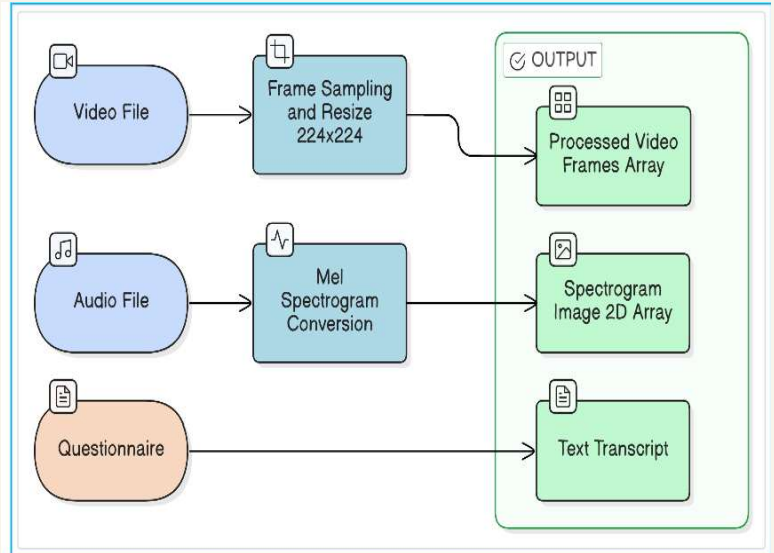


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MODULES

Data Preprocessing & Input Block

- Video Processing: Video clips are sampled into a sequence of individual frames, which are then resized and normalized. This prepares the data for the 3D-CNN.
- Audio Processing: The audio track is converted into Mel Spectrograms, creating a 2D "image" of the sound that a 2D-CNN can analyze.
- Text Processing: Text input is taken through a questionnaire.

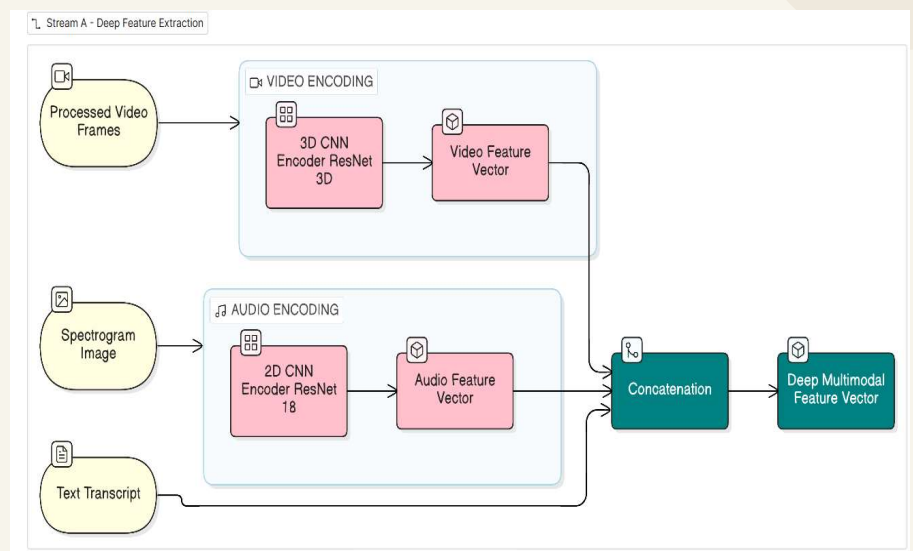


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MODULES

Stream A: Deep Feature Extraction

- Video Encoder (3D-CNN): A ResNet-3D architecture processes the sequence of video frames to capture spatio-temporal features like the dynamics of a facial expression.
- Audio Encoder (2D-CNN): A ResNet-18 processes the Mel Spectrograms to learn rich acoustic patterns related to tone and vocal quality.

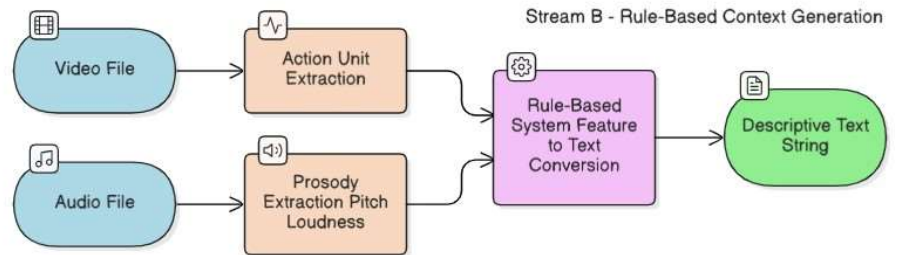


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MODULES

Stream B: Rule-Based Context Generation

- **High-Level Feature Extraction:** This module extracts specific, interpretable features, such as Facial Action Units (AUs) from the video and prosodic features like loudness and pitch from the audio.
- **Textual Conversion:** A rule-based system then converts these numerical features into a descriptive text string, such as "The speaker has a loud volume. Facial cues include raised cheeks.".

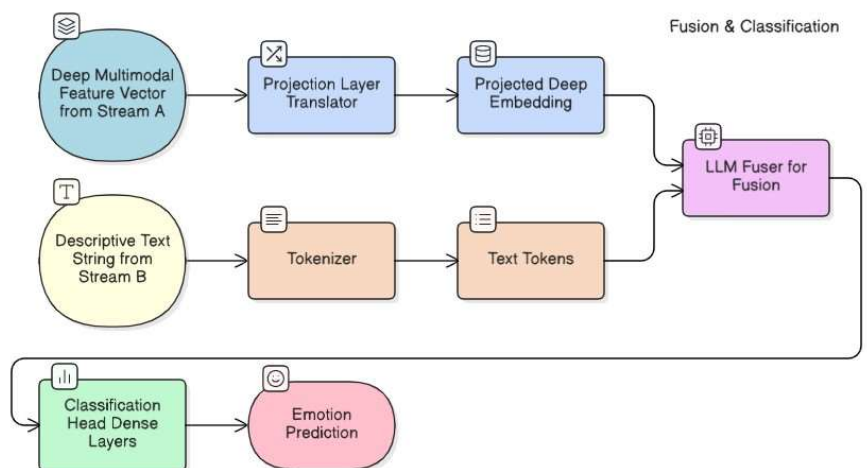


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MODULE S

Fusion Core & Classification

- **Projection Layer:** This trainable linear layer acts as a "translator," mapping the numerical vector from the Deep Feature Stream into the LLM's embedding space.
- **LLM Fuser (distilRoBERTa):** The final LLM receives the projected deep features, the rule-based descriptive text and text from questionnaire as input. Its self-attention mechanism weighs the importance of both quantitative and qualitative information to form a holistic understanding.
- **Final Classification:** The output is passed to a classification head which makes the final prediction.



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ROLES ASSIGNED

- Yuhan John - Design and implement the audio stream in the deep feature stream.
- Savio Tomy - Work on the deep feature video stream as well as the fusion layer.
- V Karthik - Implement the Mediapipe pipeline to process the video files and extract the blendshape scores and create text description along with real-time implementation.
- Volwin Shaji - Implement the openSMILE pipeline to extract specified audio features (loudness, pitch etc.) from datasets and create the text description, also work on real-time implementation.

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HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

- Processor: Intel i7 / Ryzen 7 (i5
- minimum) RAM: 16 GB (8 GB minimum)
- GPU: NVIDIA RTX 3060 or higher (8GB+ VRAM
- recommended) Storage: 512 GB SSD
- Input Devices: Webcam & Microphone

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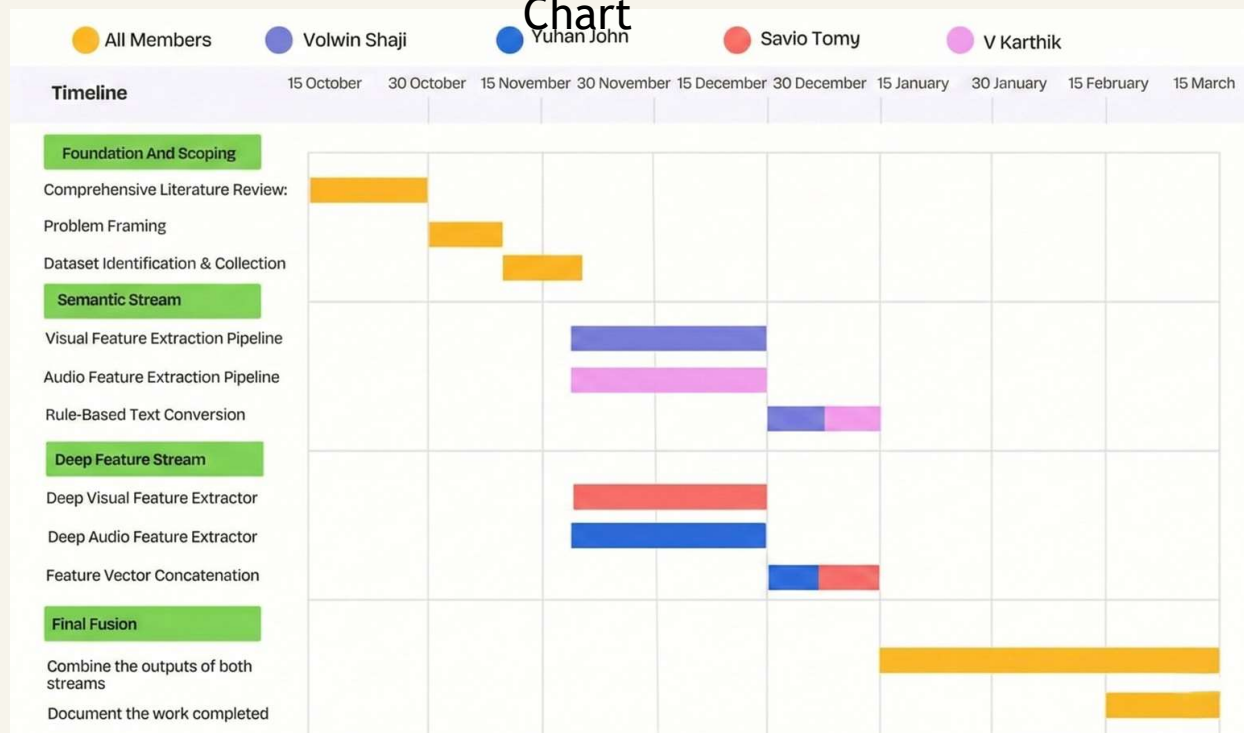
HARDWARE AND SOFTWARE REQUIREMENTS

Software:

- OS: Windows / Ubuntu / macOS
- Language: Python 3.9+
- Frameworks: PyTorch / TensorFlow
- Libraries: OpenCV, Librosa, NumPy,
- Pandas LLM Support: HuggingFace
- Transformers Tools: VS Code / Kaggle Notebook
- GPU Support: CUDA Toolkit

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Gantt Chart



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RESULTS

--- Sample / Emotion: Angry ---

Video:



Visual Analysis

Detected Action Units (Intensity > 0.1):

AU01	0.18
AU05	0.73
AU15	0.18
AU17	0.36
AU23	0.36
AU25	0.64
AU26	0.36

Name: 2114, dtype: float64

Generated Visual Text: Facial features observed: Raises upper eyelid, Parts lips slightly.

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RESULTS

Audio Analysis

Extracted Audio Features:

equivalentSoundLevel_dBp	-34.669998
F0semitoneFrom27.5Hz_sma3nz_stddevNorm	0.160000
F0semitoneFrom27.5Hz_sma3nz_pctlrange0-2	9.820000
spectralFlux_sma3_amean	0.190000
mfcc1_sma3_amean	7.510000
mfcc2_sma3_amean	-1.810000
mfcc3_sma3_amean	0.350000
mfcc4_sma3_amean	-11.570000

Name: 2114, dtype: float32

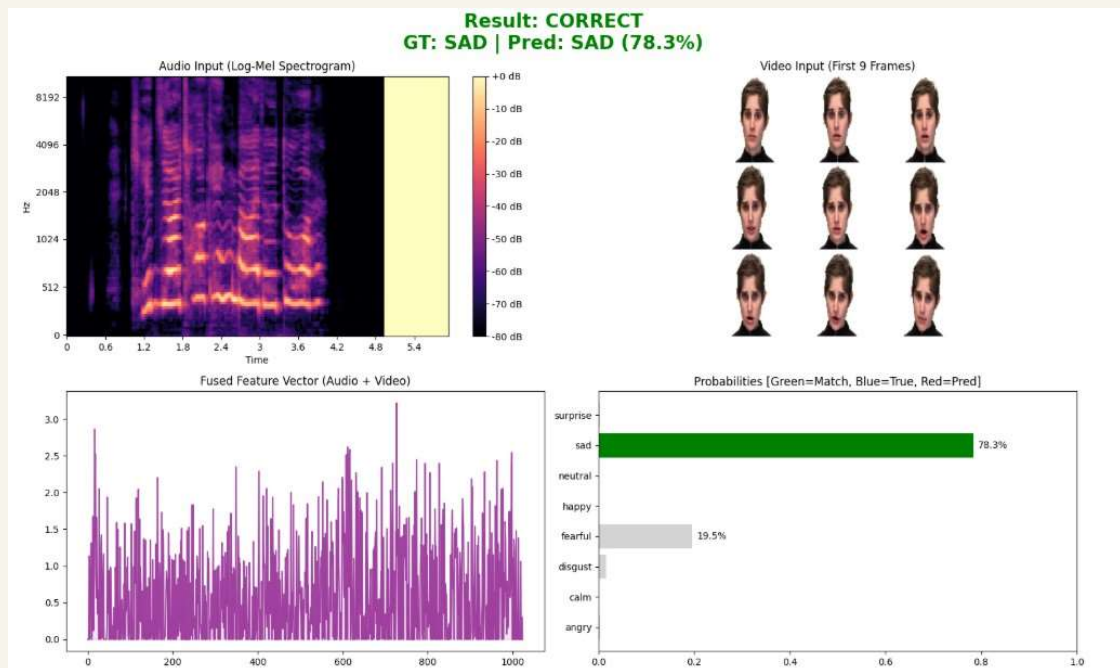
Generated Audio Text: Moderate volume. Relatively stable pitch.

Final Combined Prompt

Audio cues: Moderate volume. Relatively stable pitch. Visual cues: Facial features observed: Raises upper eyelid, Parts lips slightly.

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RESULTS



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RESULTS

```

Combined Feature Vector Shape: torch.Size([4, 1024])
Prediction Output Shape: torch.Size([4, 8])
Training on Actors: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
Validating on Actors: [21, 22, 23, 24]
Training on 2036 samples, Validating on 416 samples.
Starting training on cuda...

Results: Train Loss: 0.1717, Train Acc: 95.38% | Val Loss: 0.9362, Val Acc: 71.63%
-----
✅ Model Saved (New Best Validation Loss)
    
```

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TRAINING RESULTS &

PERFORMANCE Performance Metrics • Achieved 82.93% accuracy on RAVDESS validation set (Actors 21-24)

- F1-Score: 0.8210 - consistent performance across all emotion classes

Strong Performers (F1 > 0.85):

- ✓ Angry: 96.88% recall, 86.11% precision → F1: 0.9118
- ✓ Happy: 82.81% recall, 96.36% precision → F1: 0.8908
- ✓ Calm: 84.38% recall, 91.53% precision → F1: 0.8780
- ✓ Surprise: 93.75% recall, 76.92% precision → F1: 0.8451
- ✓ Disgust: 90.62% recall, 78.38% precision → F1: 0.8406

Moderate Performers (F1 > 0.70):

- ≈ Sad: 84.38% recall, 73.97% precision → F1: 0.7883
- ≈ Fearful: 62.50% recall, 86.96% precision → F1: 0.7273
- ≈ Neutral: 71.88% recall, 65.71% precision → F1: 0.6866

2
3

RESULTS

CLASSIFICATION METRICS

Overall Performance:

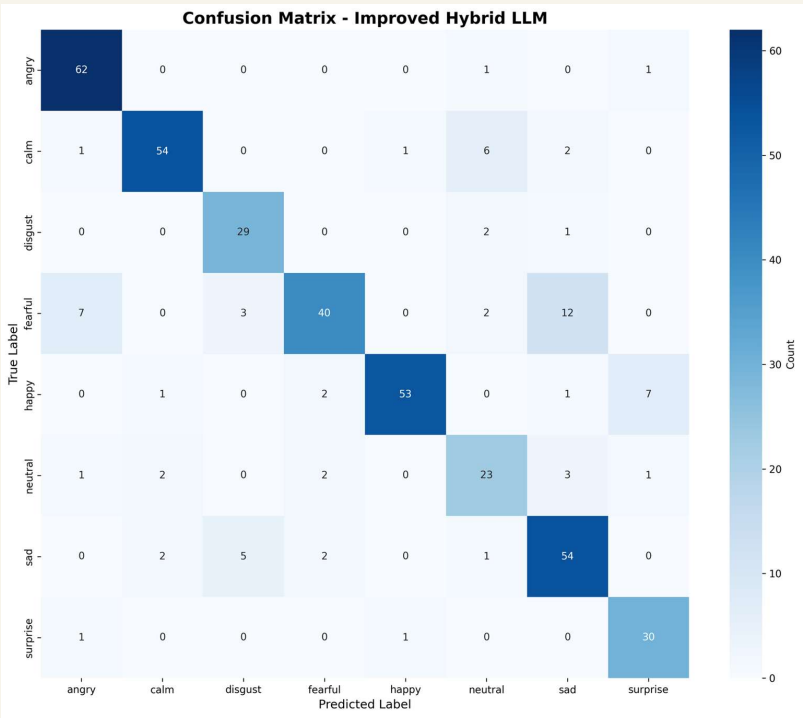
Accuracy: 82.93%
F1-Score (Macro): 0.8210
F1-Score (Weighted): 0.8280

Detailed Classification Report:

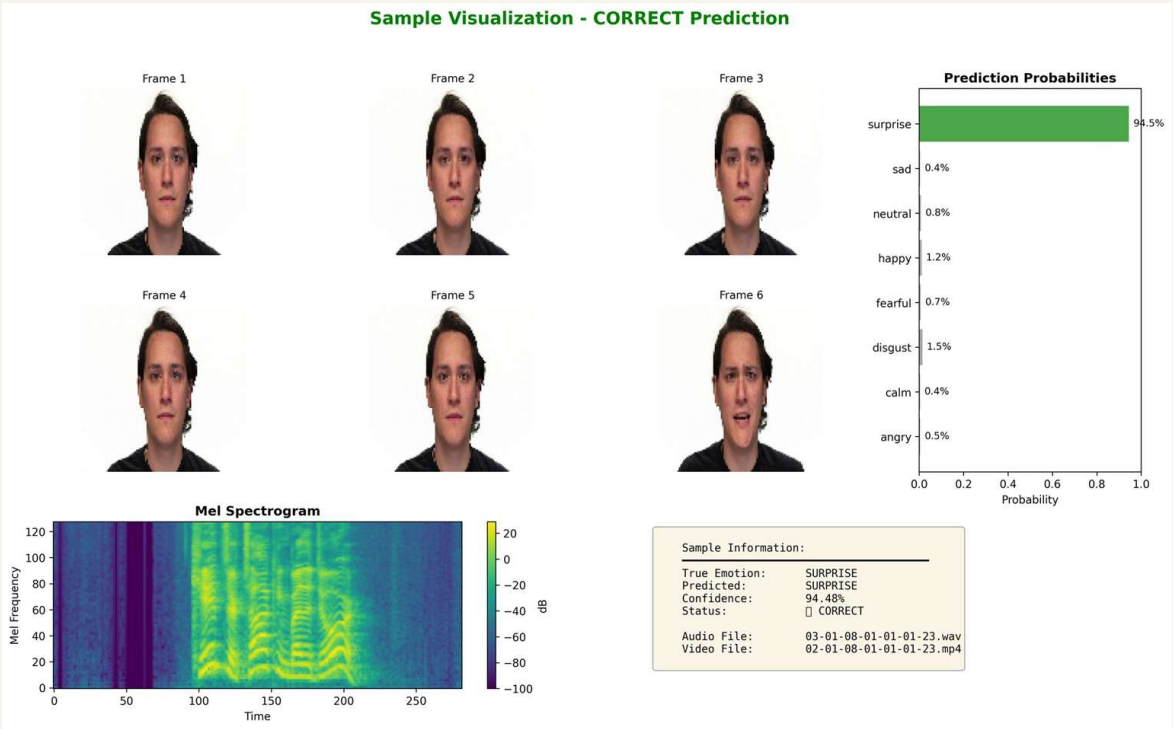
	precision	recall	f1-score	support
angry	0.8611	0.9688	0.9118	64
calm	0.9153	0.8438	0.8780	64
disgust	0.7838	0.9062	0.8406	32
fearful	0.8696	0.6250	0.7273	64
happy	0.9636	0.8281	0.8908	64
neutral	0.6571	0.7188	0.6866	32
sad	0.7397	0.8438	0.7883	64
surprise	0.7692	0.9375	0.8451	32
accuracy			0.8293	416
macro avg	0.8199	0.8340	0.8210	416
weighted avg	0.8391	0.8293	0.8280	416

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RESULTS



RESULTS



ASSUMPTIONS

- Users provide clear video and audio recordings with minimal background noise.
- Facial expressions are clearly visible for accurate Action Unit extraction. Audio quality is sufficient for reliable prosody analysis.
- Users respond to the questionnaire honestly and completely.
- The system operates under adequate computational resources (GPU- enabled environment).
- Emotions expressed belong to the predefined emotion categories used for training.

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RISKS

- **Multimodal Complexity:** Combining audio, video, and text may introduce synchronization challenges, leading to inconsistent emotion recognition.
- **Accuracy Issues:** Fusion models may still produce false predictions (misclassifying emotions) under noisy or ambiguous conditions.
- **Real-Time Constraints:** Processing three input modalities simultaneously may cause delays or require high computational resources.
- **Scalability & Maintenance:** Continuous updates and retraining may be necessary to maintain accuracy as new datasets and use cases emerge.

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CONCLUSION

- The project shows that combining audio, visual, and linguistic data leads to significantly more accurate emotion recognition than using any single type of input.
- By analyzing facial expressions, voice tone, and spoken words together, the system captures deeper emotional context and handles complex real-world scenarios better.
- Deep learning models were used to extract and fuse features from each modality, resulting in a robust and integrated emotion classification system.

The system has practical applications in areas such as mental health monitoring, emotion-aware virtual assistants, and human-computer

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FUTURE SCOPE

- Real-World System Deployment
- Develop a real-time emotion recognition application.
- Accept live video and audio input through webcam and microphone.
- Implement real-time preprocessing pipelines for low latency
- performance. Deploy trained model using optimized inference frameworks.

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Achievements

- **Recipient of IndiaAI Fellowship Award**

Honoured with the prestigious **IndiaAI Fellowship Award of Rs 1 Lakh** in recognition of excellence, innovation, and impactful contributions to Artificial Intelligence research.

- **Research Paper – Under Review**

A research work titled “**A Unified Multimodal Deep Learning Framework for Emotion Recognition via Audio-Visual-Textual Feature Fusion**” communicated to Journal of Science & Technology Asia – Scopus Indexed.

- **Industry-Supported Project**

Successfully developed an industry-supported project in collaboration with **BaseOne Innovations Pvt Ltd, Kakkanad** integrating real-world requirements with advanced AI methodologies.

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Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.

PO2: Problem Analysis Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

- PO3: Design/Development of Solutions** Design creative solutions for complex engineering problems and design/develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
- PO4: Conduct Investigations of Complex Problems** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.
- PO5: Engineering Tool Usage** Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling while recognizing their limitations to solve complex engineering problems.
- PO6: The Engineer and The World** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- PO7: Ethics** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.
- PO8: Individual and Collaborative Team Work** Function effectively as an individual and as a member or leader in diverse and multidisciplinary teams.
- PO9: Communication** Communicate effectively and inclusively within the engineering community and society at large.
- PO10: Project Management and Finance** Apply knowledge and understanding of engineering management principles and economic decision-making to manage projects in multidisciplinary environments.
- PO11: Life-Long Learning** Recognize the need for and have the preparation and ability for independent and life-long learning, adaptability to new and emerging technologies and critical thinking.

Programme Specific Outcomes (PSO)

PSO1: Computer Science Specific Skills

Ability to identify, analyze, and design solutions for complex engineering problems by understanding the core principles of computer science.

PSO2: Programming and Software Development Skills

Ability to design algorithms and apply software development practices to deliver quality software products.

PSO3: Professional Skills

Ability to apply computer science fundamentals in research and develop innovative solutions to meet societal needs.

Course Outcomes (CO)

After completion of the course the student will be able to

SL.NO	DESCRIPTION
CO1	Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).
CO2	Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).
CO3	Function effectively as an individual and as a leader in diverse teams (Cognitive knowledge level: Apply).
CO4	Plan and execute tasks utilizing available resources within timelines following ethical and professional norms (Cognitive knowledge level: Apply).
CO5	Identify technology or research gaps and propose innovative solutions (Cognitive knowledge level: Analyze).
CO6	Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1-PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1-PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.

Mapping	Level	Justification
101003/ CS822U.CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2-PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3-PO9	Medium	Requires effective communication among team members during collaboration.

Mapping	Level	Justification
101003/ CS822U.CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4-PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.

Mapping	Level	Justification
101003/ CS822U.CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5-PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5-PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6-PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6-PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.

Mapping	Level	Justification
101003/ CS822U.CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.